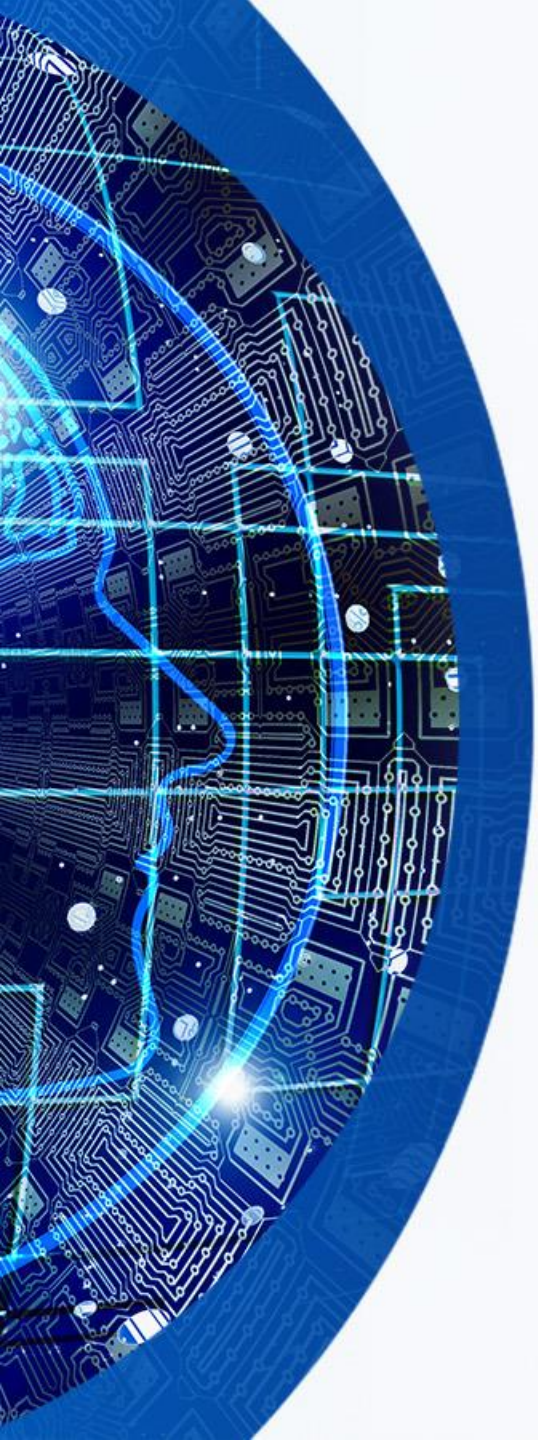




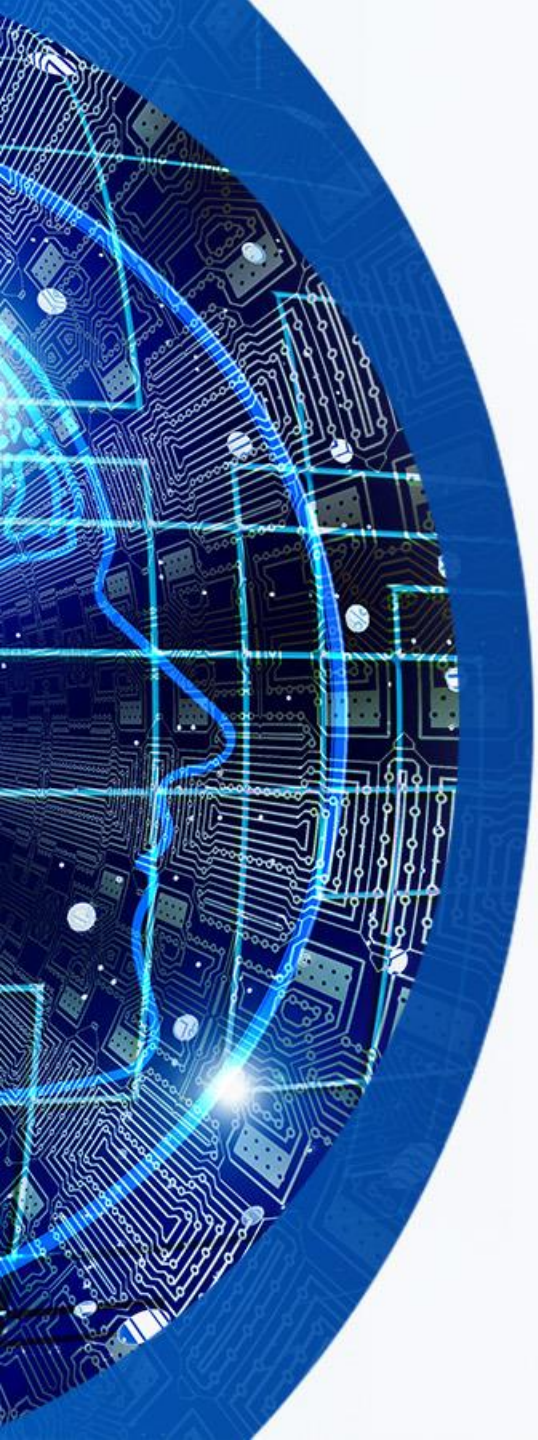
Natural Language Processing

Arabic NLP



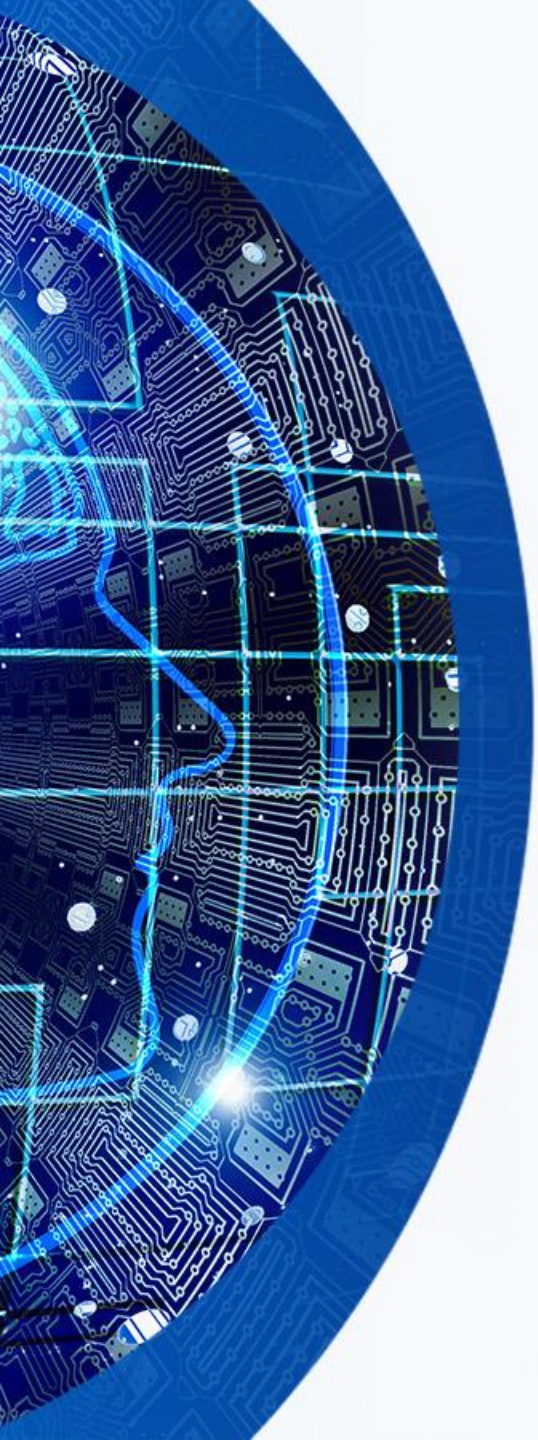
Outlines

- Introduction
- Challenges
 - Arabic Orthography
 - Arabic morphology
 - Syntax is intricate



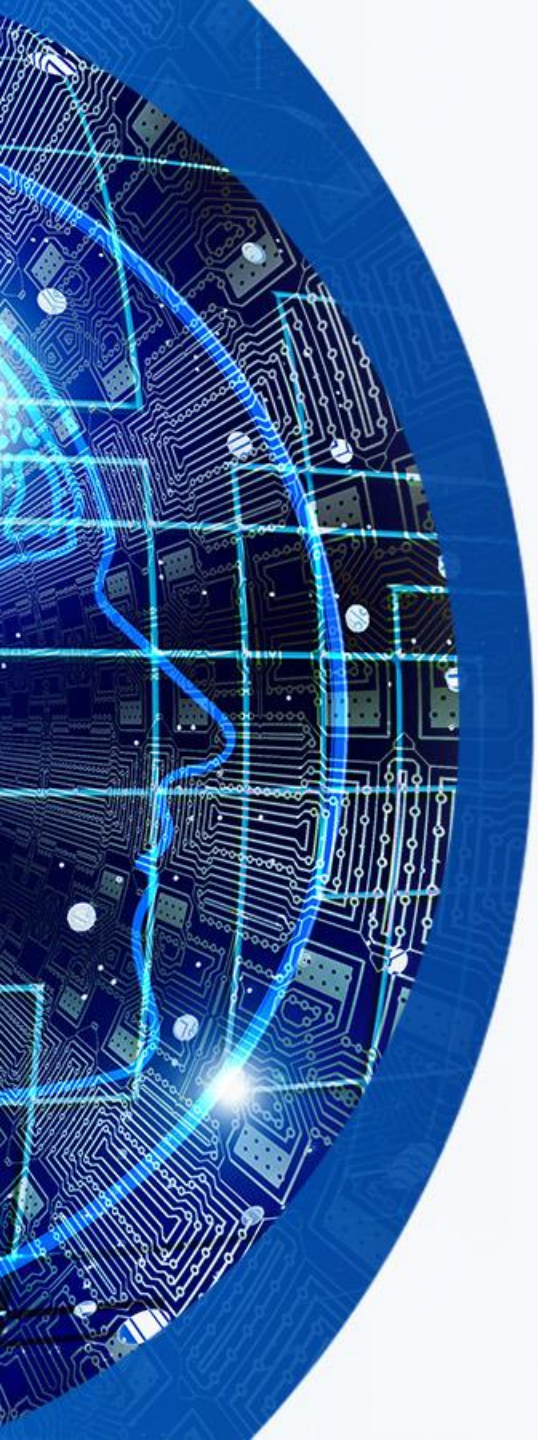
Introduction

- Natural language processing (NLP) is a domain of computer science that aims at facilitating communication between machines (computers that understand machine language or programming language) and human beings (who communicate and understand natural languages like English, Arabic and Chinese etc.)
- NLP is very important as it makes a huge impact on our daily lives.
- Many applications these days use concepts from NLP.



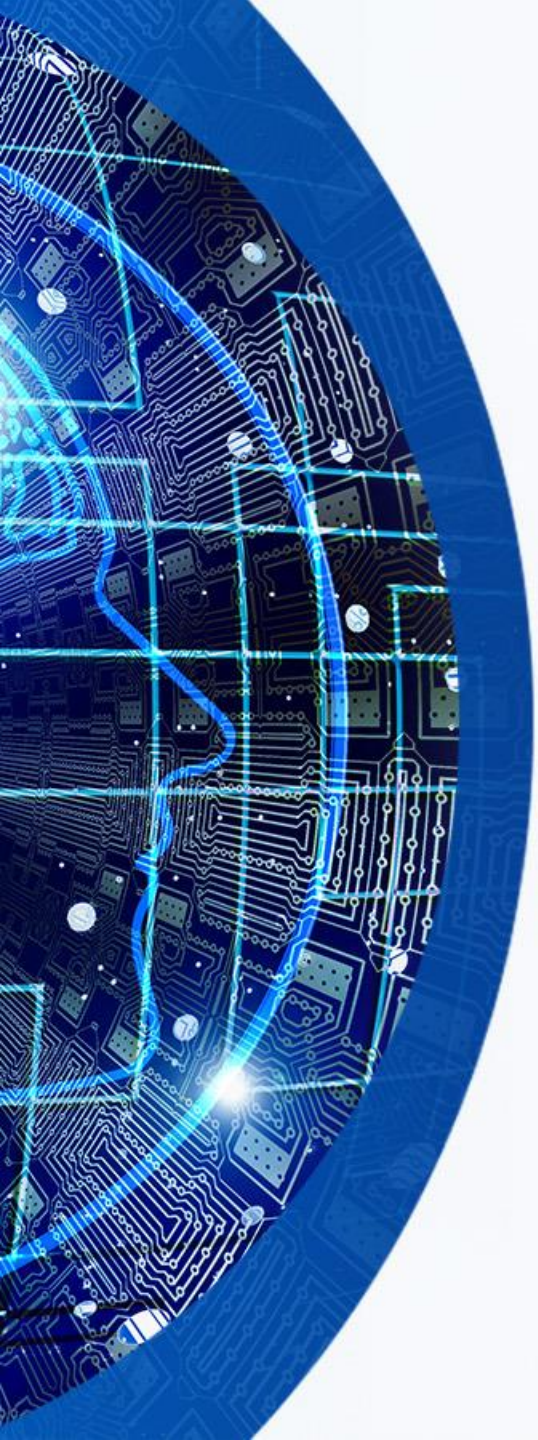
Introduction

- Arabic is the first language of the Arab world countries, which has significant importance worldwide.
- Arabic is an uncommonly rich language which is related to another linguistic family, particularly Semitic vernaculars, which is different from the Indo-European lingos talked in the West.
- Arabic is interesting and any person with a slight knowledge of Arabic can read and understand a text **written fourteen centuries ago.**



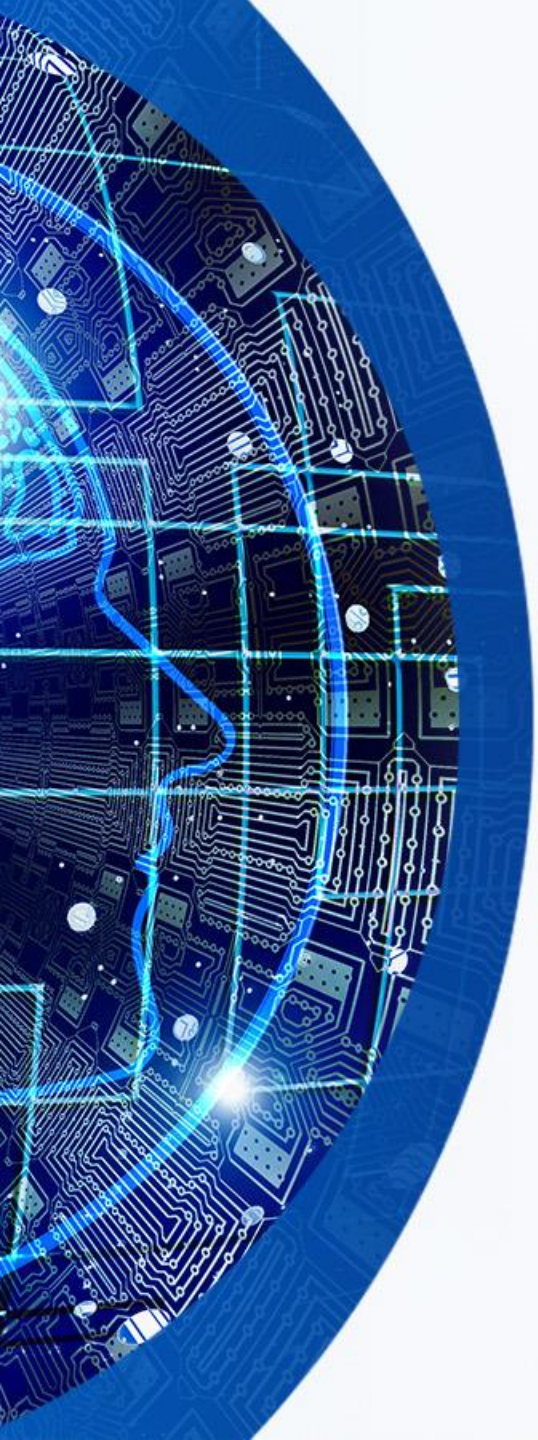
Introduction

- Arabic, as a language or dialect is exceedingly derivational and inflectional, and there are **no rules for emphasis**.
- Truly, there are **principles**; however, there are no firm rules
- Arabic language has a rich and complex grammatical structure.
- For instance, a noun and its modifiers **need to agree** in number, gender, case, and definiteness.
- Moreover, in Arabic, there are advancements that really mean, “**Mother of**” or “**father of**” to show ownership, a trademark, or a property, and use gendered pronouns; it has **no fair-minded pronouns**



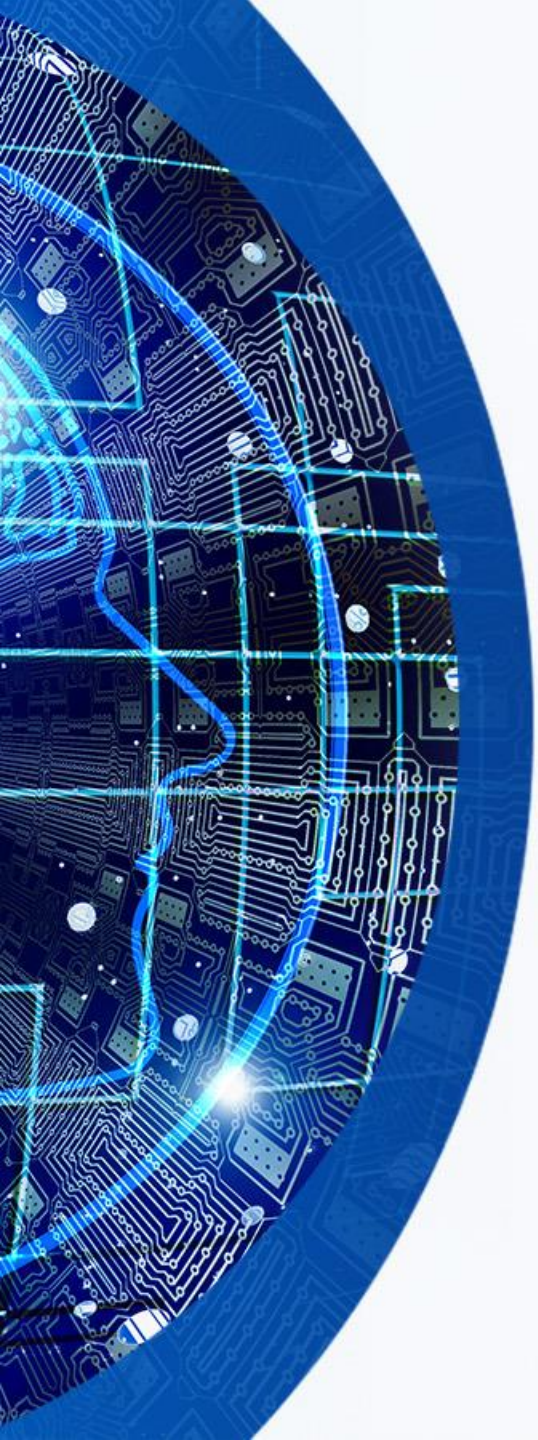
Introduction

- Arabic sentences can be nominal (subject–verb), or verbal (verb–subject) **with free order**; however, English sentences are fundamentally in the (subject–verb) order.
- The free order property of the Arabic language presents a **crucial challenge** for some Arabic NLP applications.



Introduction

- Three types characterize Arabic:
 - **Classical (Traditional or Quranic) Arabic,**
 - **Modern Standard Arabic**
 - **Dialect Arabic.**
- Arabic language takes these forms in light of three key parameters including morphology, syntax and lexical mixes.
- Classical Arabic is primarily use in Arabic speaking countries.
- **Diacritic** marks (also known as “**Tashkil**” or short vowels) are commonly used within Classical Arabic as **phonetic** guides to show the correct **pronunciation**.
- On the contrary, diacritics are considered optional in most other Arabic writing



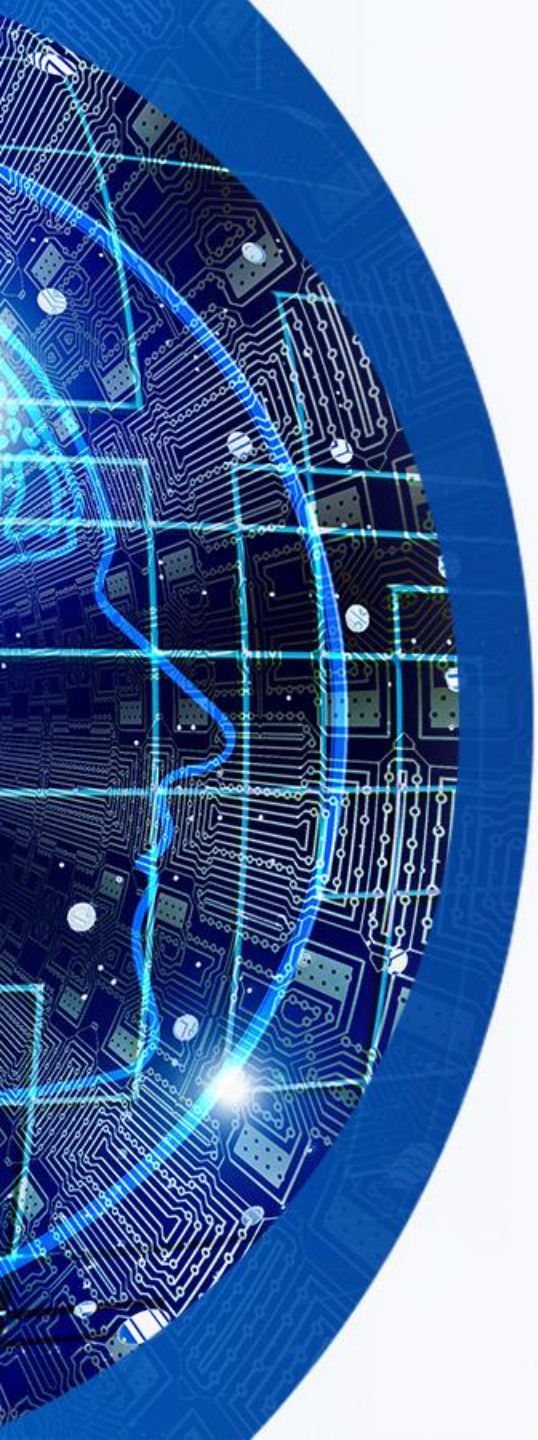
Introduction

- Modern Standard Arabic (MSA) is used for TV, newspapers, poetry and in books.
- Arabic Courses at the Arab Academy are also taught in the Modern Standard form.
- The MSA can be transformed to adapt to new words that need to be created because of science or technology.
- However, the written Arabic script has seen no change in the alphabet, spelling or vocabulary in at least four millenniums.

Introduction

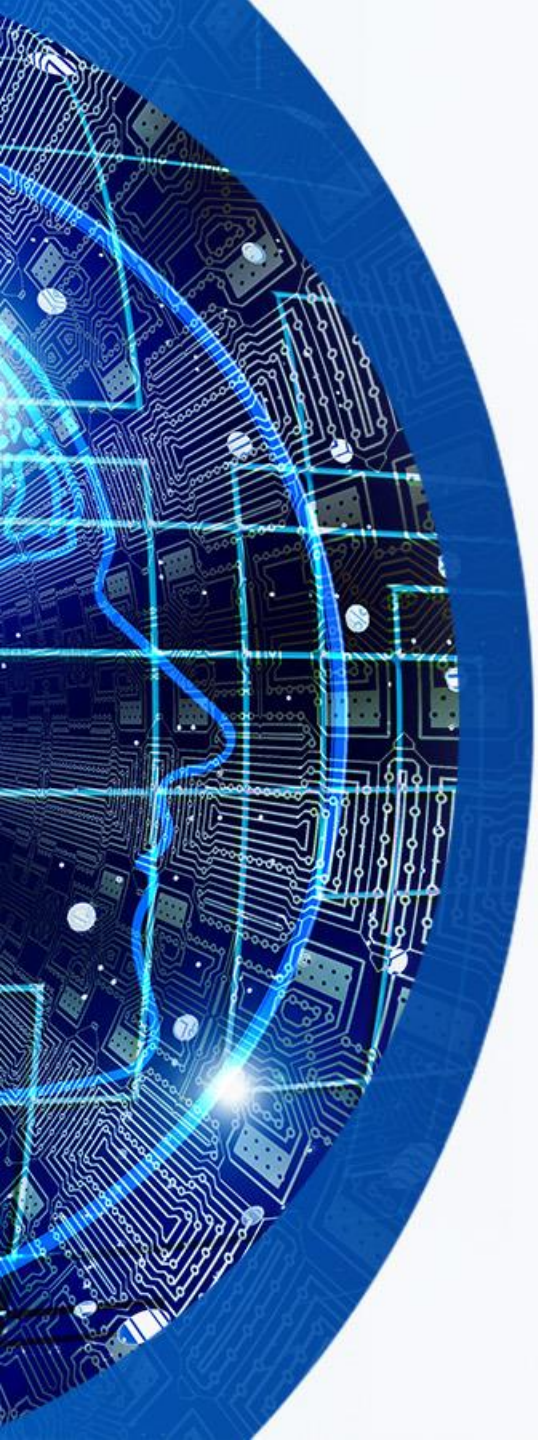
Diagram illustrating the components of Arabic calligraphy for the phrase "الخليفة أبو بكر الصديق" (The Caliph Abu Bakr al-Siddiq). The components are labeled with red circles and arrows:

- Dots: Points to the dots on the letter 'ي' (Ya).
- Loop: Points to the loop in the letter 'ص' (Sin).
- Overlap: Points to the overlap in the letter 'ب' (Ba).
- Diacritic: Points to the diacritic 'ة' (Ta) on the letter 'ب' (Ba).
- Dot: Points to the dot on the letter 'ف' (Fay).
- Dots: Points to the dots on the letter 'ي' (Ya).
- Ligature: Points to the ligature between the letters 'ف' (Fay) and 'ي' (Ya).



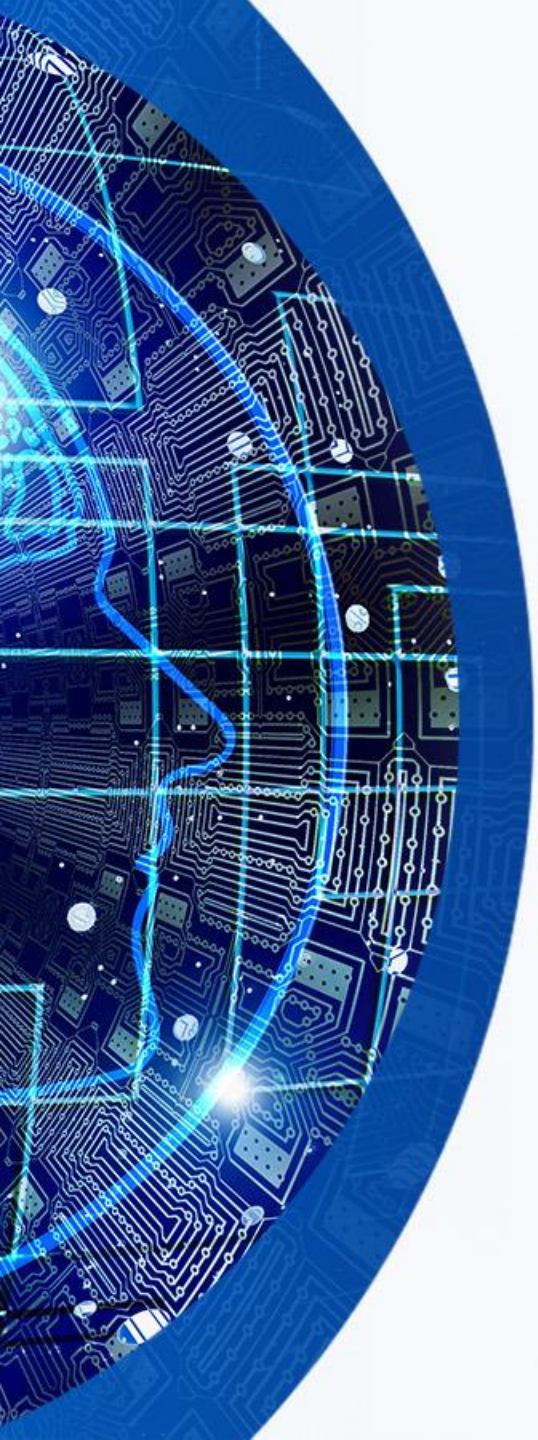
Introduction

- Dialect Arabic or “colloquial Arabic” is casually utilized daily by Arabs.
- It is grouped into Mesopotamian Arabic, Arabian Peninsula Arabic, Syro Palestinian Arabic, Egyptian and Maghrebi Arabic.
- Arabic dialect generally used, mostly written, by Internet clients and social media; varies from locale to area is Dialect Arabic.
- In vernacular Arabic, portions of the words are acquired from MSA.
- Some studies presented a hybrid pre-processing approach that has the ability to convert paraphrases of Egyptian dialectal input into MSA such that the available NLP tools can be applied to the converted text, as well as its enhanced version, worked on Sentiment Analysis on the data containing different Arabic Dialects.



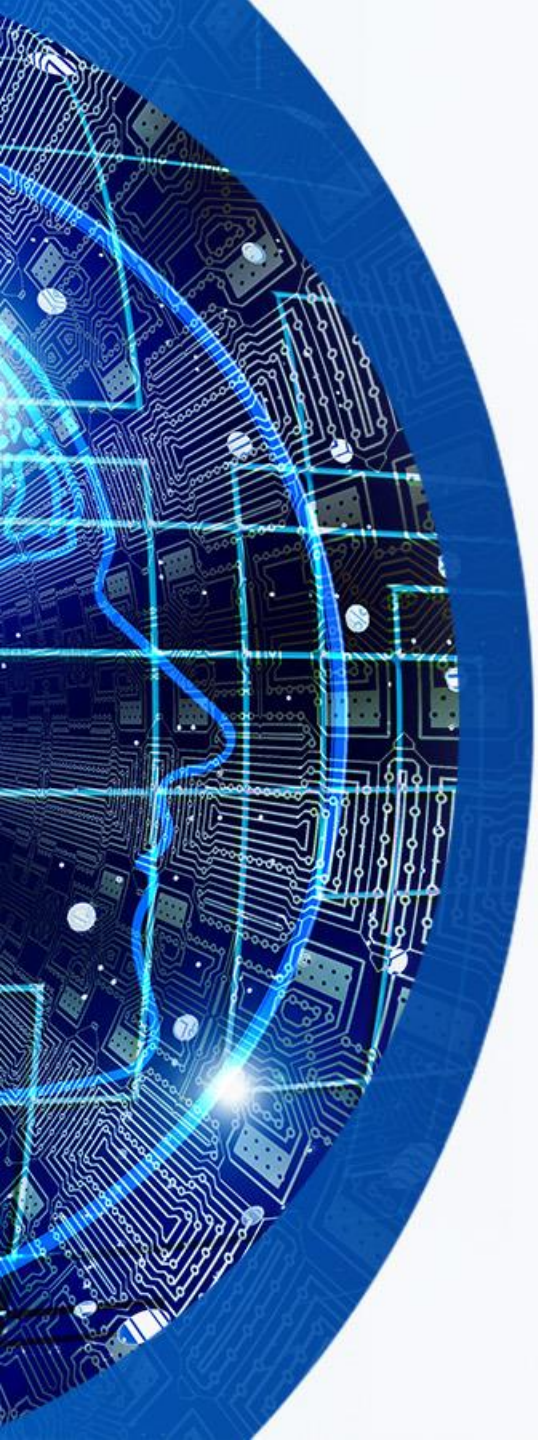
Challenges

- The Arabic is an extremely bended tongue, with unique sound, especially when pronounce the letters “ (ض ḍād), “ (ظ Tha'a), and “ (غ Ghain).
- Arabic grammar has a rich morphology and intricate sentence structure and grammarians have described it as the language of ḍād (“ لغة الضاد “)
- Arabic has a greatly rich morphology depicted by a mix of templatic and affixational morphemes, complex morphological norms, and a rich part system.
- Arabic makes use of many inflections because of the appendages, which incorporate relational words and pronouns.
- Arabic morphology is perplexing because there are about 10,000 roots that are the basis for nouns and verbs. There are 120 patterns in Arabic morphology



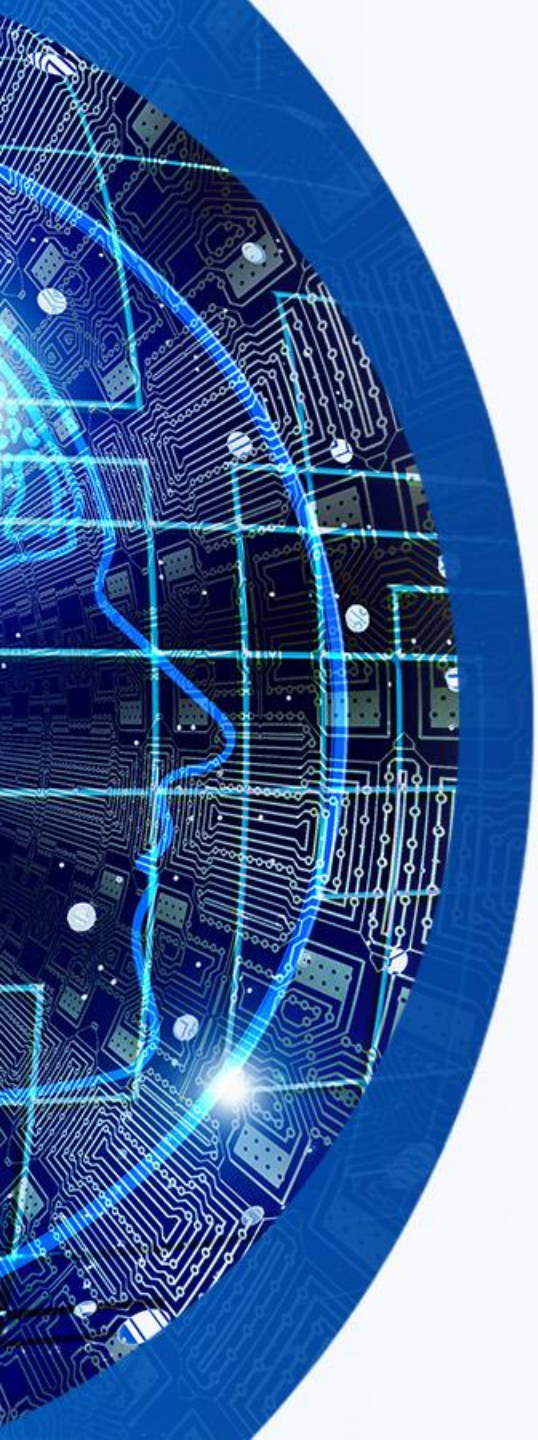
Challenges

- The word order in Arabic is variant.
- We can have a free choice of the word we want to emphasize and put it at the head of sentence.
- Generally, the syntactic analyzer parses the input tokens produced by the lexical analyzer and tries to identify the sentence structure using Arabic grammar rules.
- The relatively free word order in an Arabic sentence causes syntactic ambiguities which require investigating all the possible grammar rules as well as the agreement between constituents



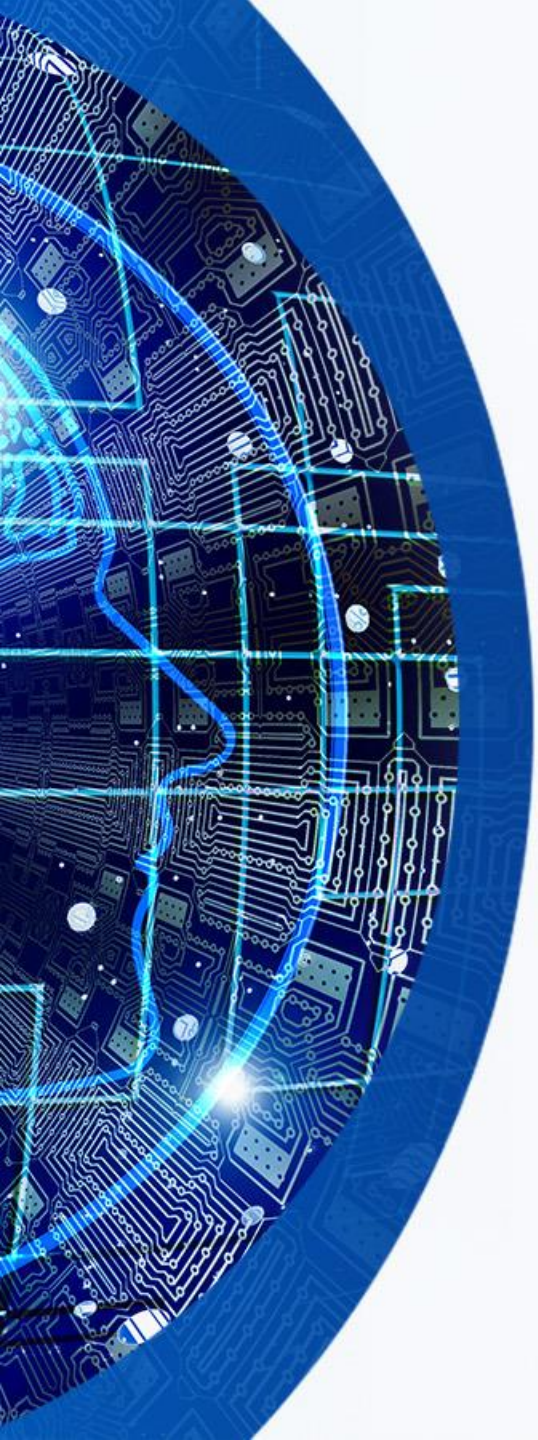
Challenges

- we will discuss the challenges of Arabic language with regard to its characteristics and their related computational problems at orthographic, morphological, and syntactic levels.
- In automating the process of analyzing Arabic sentences, there is an overlap between these levels, as they all help in making sense and meaning of words, and in disambiguating the sentence.



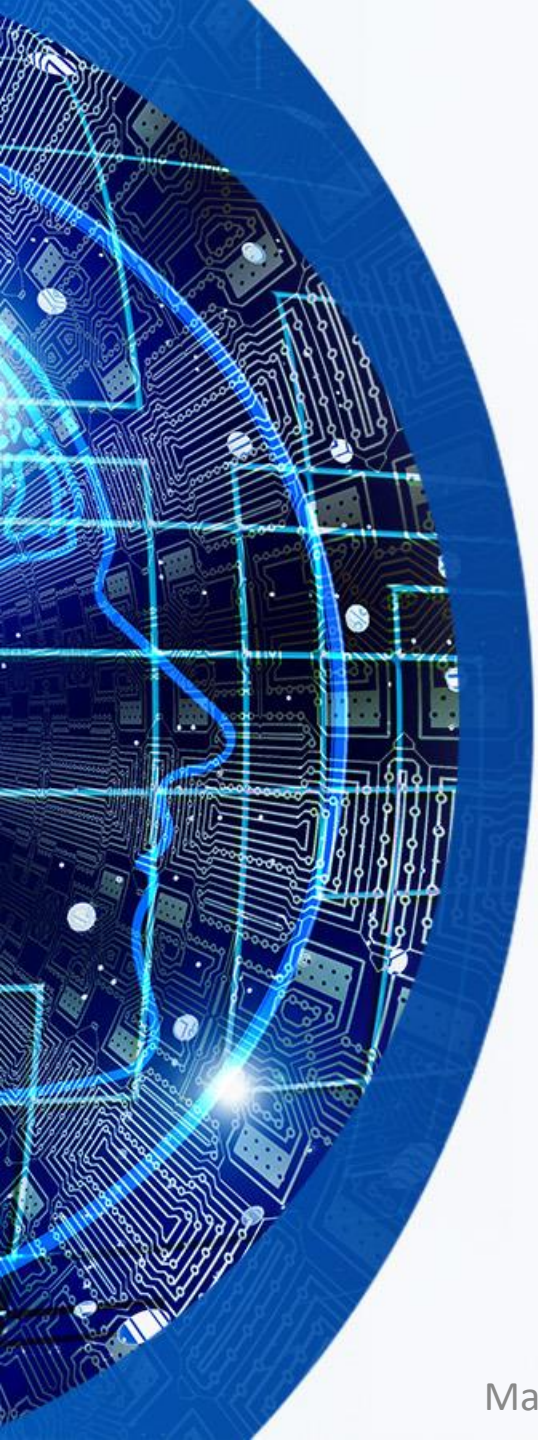
Challenges

- Arabic Orthography:
 - Diacritics,
 - Lack of consistency in orthography
 - Hamza spelling
 - Defective Verb Ambiguity
 - Nonappearance of capital letters
 - Vowels
 - Lack of uniformity in writing styles
- Arabic morphology
 - Morphology is intricate
 - Morphology declension
 - Annexation
- Syntax is intricate
 - Multi word expressions
 - Anaphora resolution
 - Ambiguous Anaphora
 - Hidden Anaphora
 - Syntactically flexible text sequence
 - Agreement



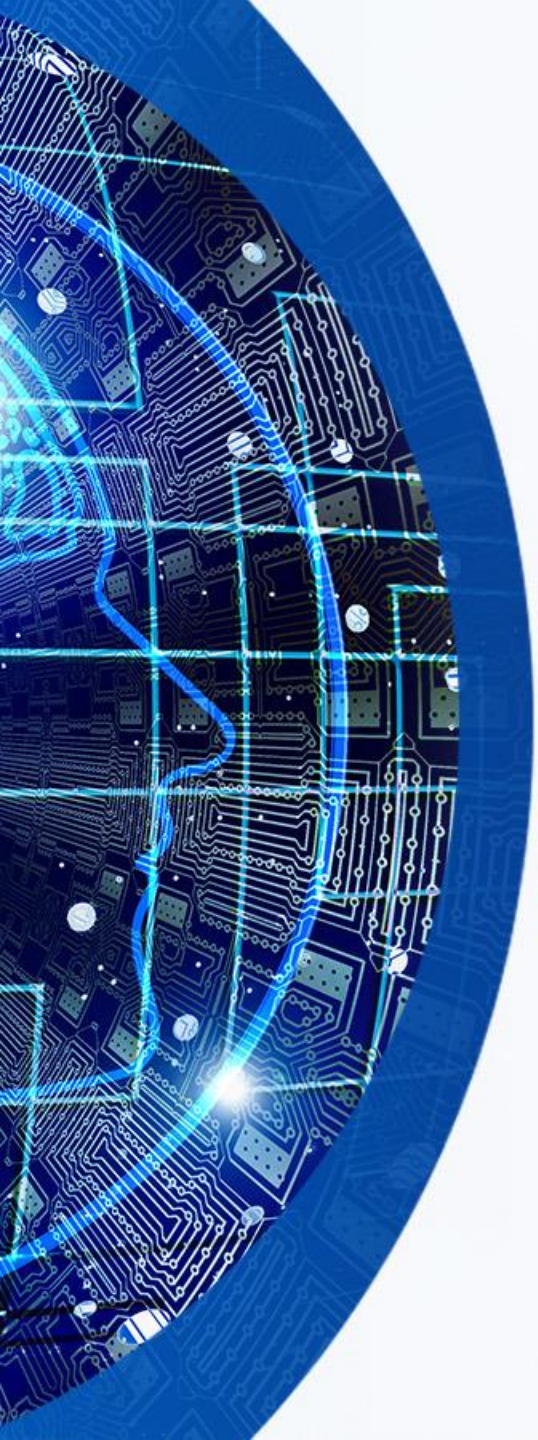
Arabic orthography

- Within the orthographic patterns of the written words, the **shape of a letter** can be changed depending on whether it is connected with a former and subsequent letter, or just connected with a former letter.
- For example, the shapes of the letter “ف” (f), i.e. “ف”, “ف”, “ف” changes depending on whether it occurs in the beginning, middle, or end of a word, respectively.
- Arabic orthography includes a set of orthographic symbols, called **diacritics**, that carry the intended pronunciation of words.



Arabic orthography

- these vowel signs are absolutely necessary in order for children and those who are not well versed in classical Arabic language to pronounce Arabic text properly.
- This helps **clarify the sense and meaning** of the word.
- The **absence of short vowels** (e.g. inner diacritics) prompts diverse sorts of ambiguities in Arabic writings (both basic and lexical) on the grounds that distinctive diacritics speak to distinctive implications.
- These ambiguities can be determined just by relevant data and a satisfactory information of the language or dialect.

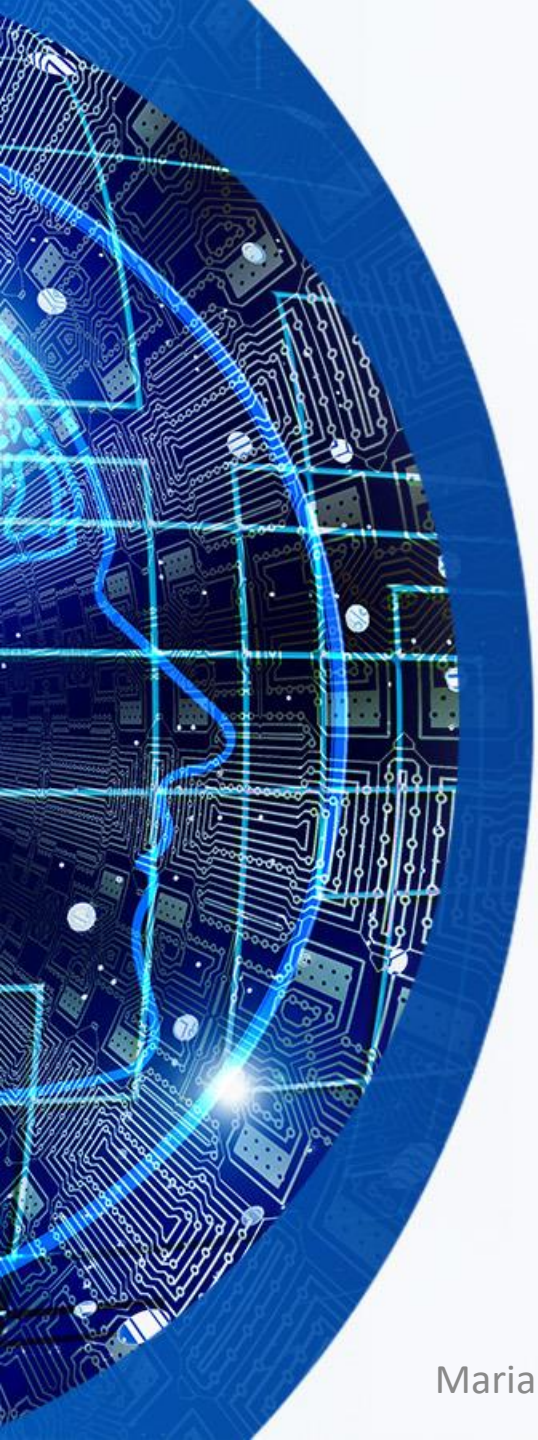


Arabic orthography

- **Contextual features** and an **adequate knowledge** of the language can only resolve these ambiguities
- For instance, the unvoweled word “كتب” (ktb):

word “كتب” (ktb), supports several alternatives such as “كَتَبَ” (he wrote, kataba), “كُتِبَ” (it was written, kutiba), “كُتُبُ” (books, kutubun), etc.

- Voweled spelling is taught to novice readers, while unvoweled spelling constitutes the standard form and is gradually imposed at later reading literacy stages.



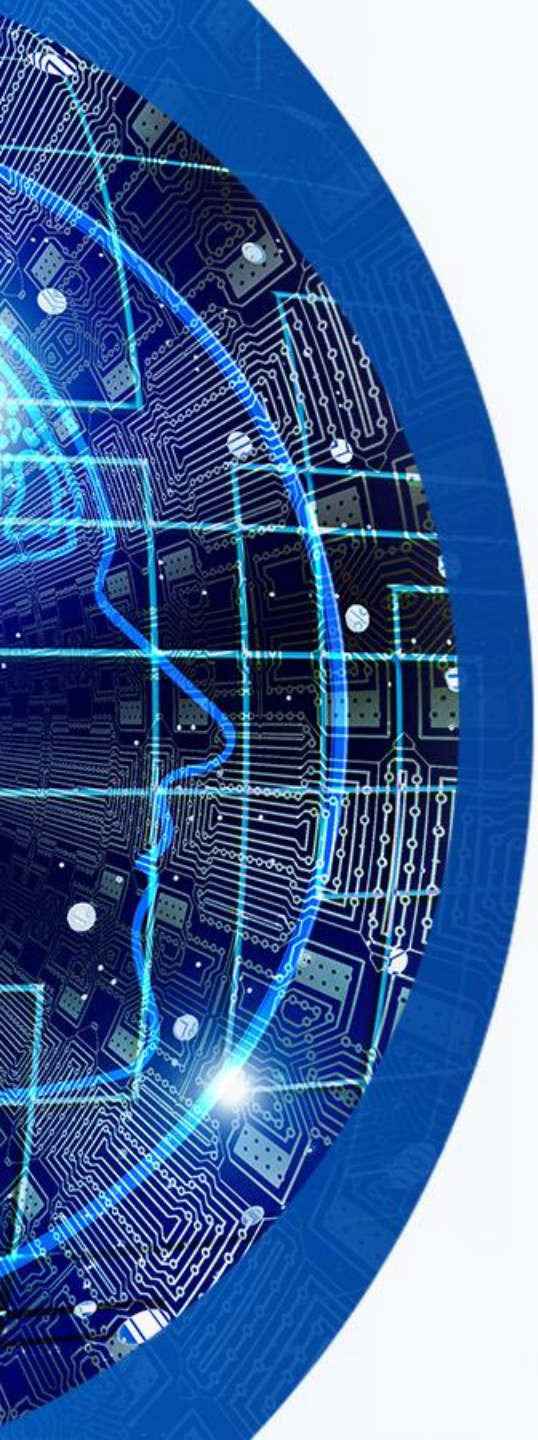
Hamza Spelling

- The most critical use of **Hamza** letter (الهمزة “ء”) brings in more challenges.
- With the very significance of Hamza being an additional letter seen at the top or bottom of the letters following the sounds of “و”, “ا”, or “ى”, i.e. “ؤ”, “أ”, or “ئ”, respectively.
- As these rules are **confusing even for native speakers**, **Hamza** is ignored most of the time while typing.
- NLP based systems should handle this assumption.

Hamza Spelling

Table 1. The Hamza diacritic is determined by its own diacritics and the preceding letter.

Example	Tashkeel of Hamza	Tashkeel of letter before Hamza	Pronunciation	Translation
سَأَلَ	Fatha	Fatha	Sa-ala	Asked
سُئِلَ	Kasra	Kasra	So-ela	Was asked
سُؤَالَ	Fatha	Dama	So-aal	Question



Defective Verb Ambiguity

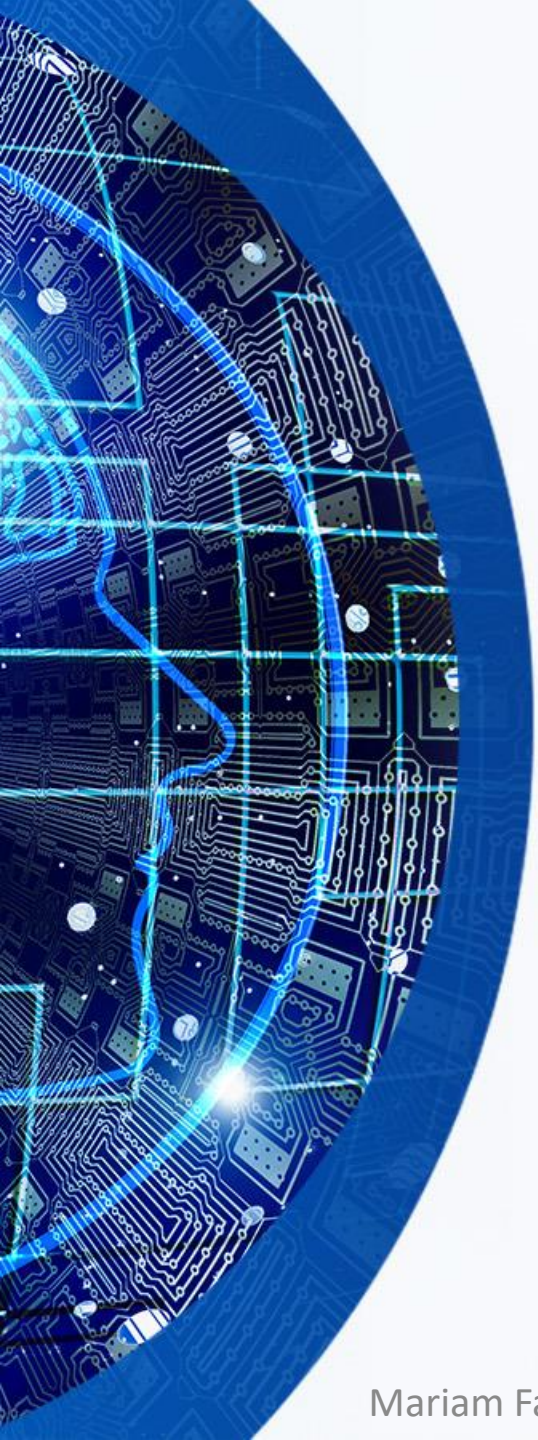
- **Defective** (weak) verb (“المعتل الفعل”) is any verb that its root has a long vowel as one of its three radicals.
- These long vowels will go through a **change** when the verb is **conjugated**.
- For example, consider the case of a **negated present** tense verb that is preceded by the apocopative particle Lam—حرف الجزم . “لم—
- In Arabic, this particle is used for negating a present tense verb form which is understood as a **negated past** form.

Defective Verb Ambiguity

- It is one of the defining features of **Modern Standard Arabic**, and is not used in **any dialects**.
- Being able to use this word properly and effectively will bring Arabic language to a higher level

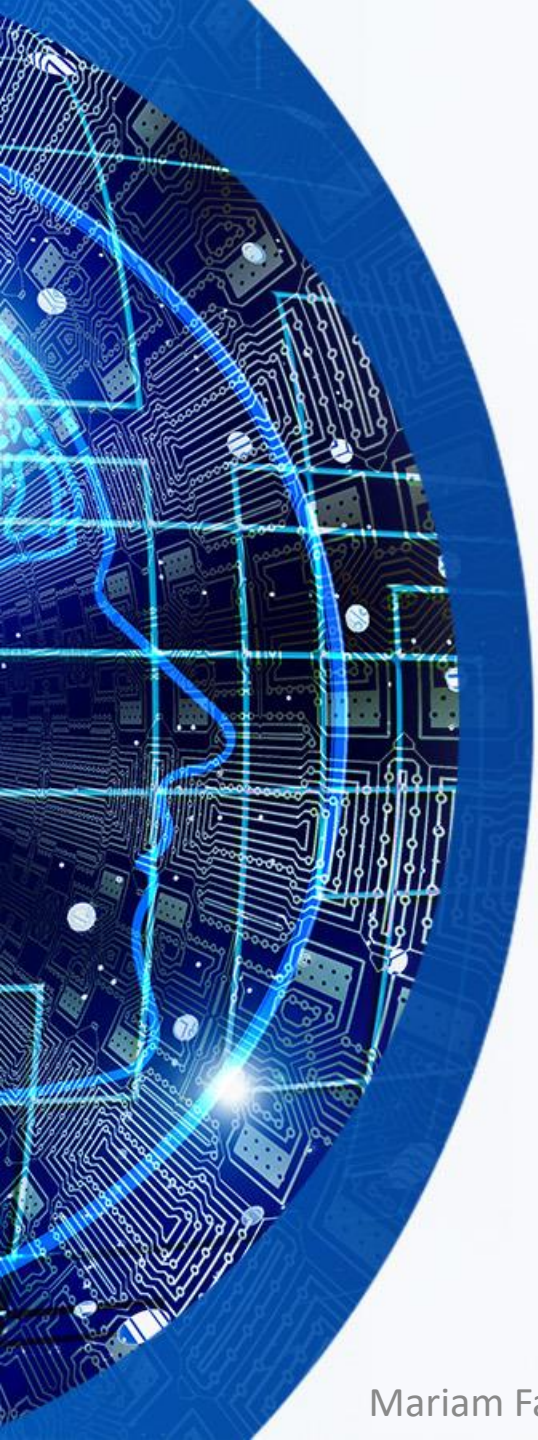
Table 2. Examples of negated past tense verb form.

Verb	Transliteration	Sentence	Change applied to the present form of the verb
دعا	Da-aa	لم يدع	Omit the last long vowel “و” and add the present tense letter “ي”
سعى	Sa-aa	لم يسع	Omit the last long vowel “ى” and add the present tense letter “ي”
صلى	Sala	لم يصل	Omit the last long vowel “ي” and add the present tense letter “ي”
زار	Zara	لم يزّر	Omit the middle long vowel “ا” and add the present tense letter “ي”



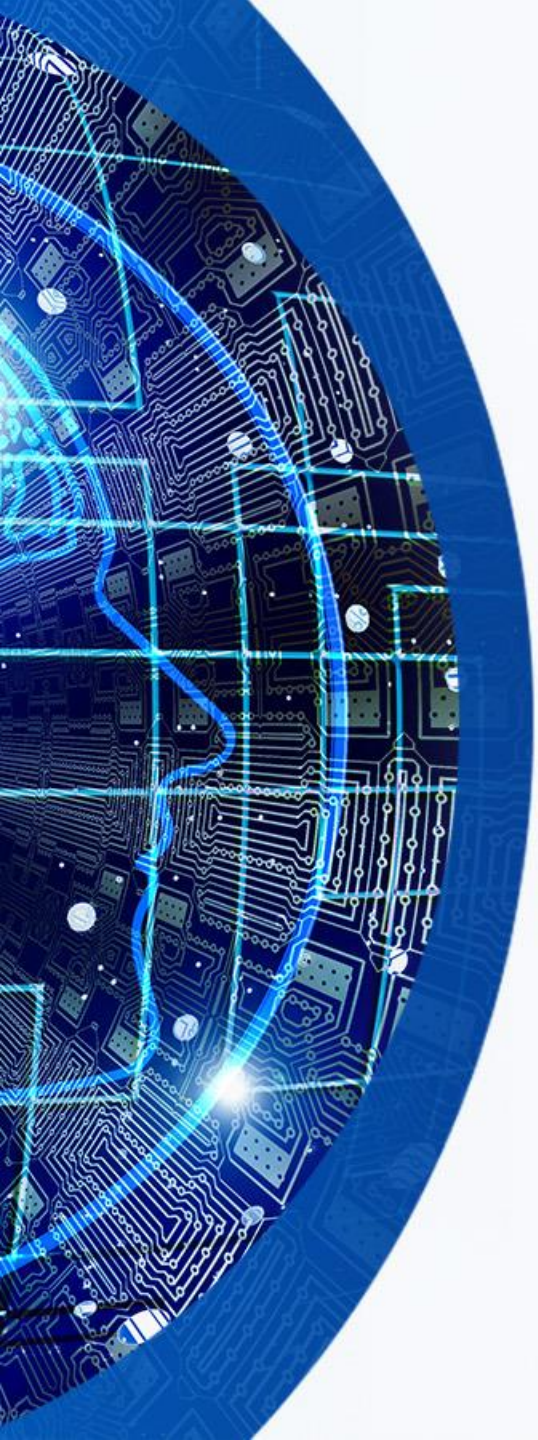
Nonappearance of capital letters

- Arabic does **not have capital letters**; this trademark speaks to an extensive hindrance for the basic task of **Named Entity Recognition** in light of the fact that in different languages, capital letters speak to a vital highlight in **distinguishing formal people, places or things**.
- Along these lines, the issue of distinguishing appropriate names is especially **troublesome for Arabic**.
- For instance, in English, capital letters are used, e.g. “**A**dam”, but no capital letter in the same name in Arabic, e.g. “آدم”



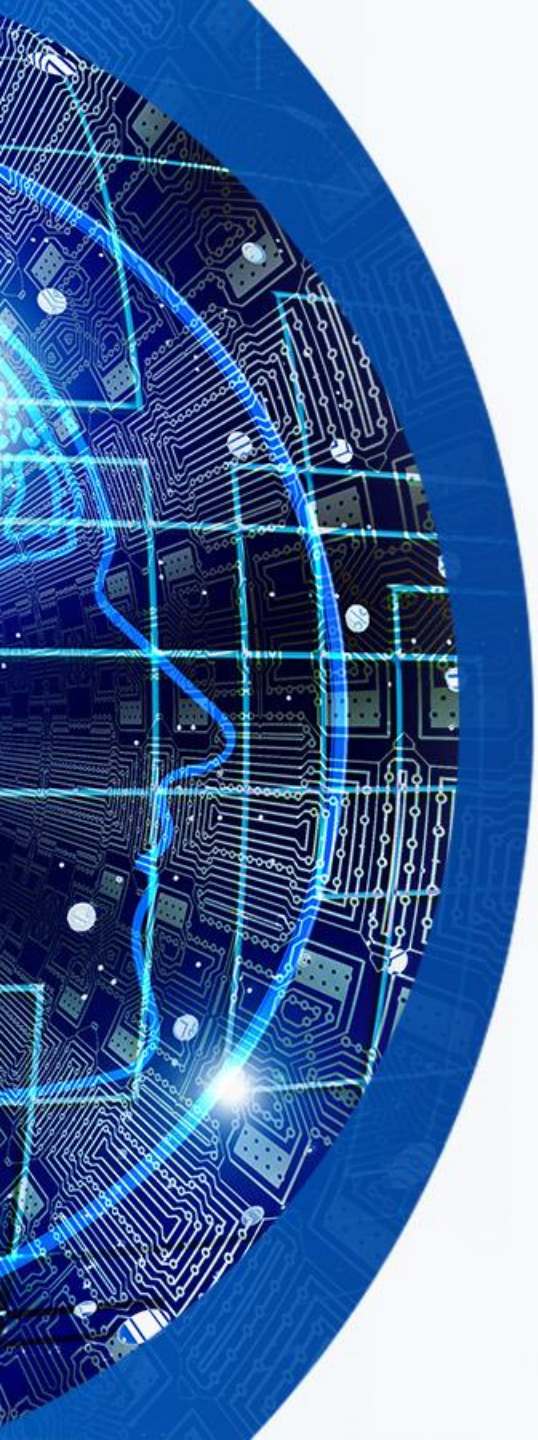
Nonappearance of capital letters

- Another reality about Arabic to consider is that the vernacular has no capital letters (e.g. for proper names: the names of people, countries, months, days of the week); therefore, cannot make usage of **acronyms**.
- This can lead to **confusion**, especially **during Information Extraction** in general and **Named Entity Recognition** in particular.
- It makes it difficult to see names of substances.
- For example, the **NE** “ المتحدة العربية الامارات ” has the acronyms **UAE** in English but not in Arabic.



Nonappearance of capital letters

- Therefore, it is common to resolve the nonappearance of capital letters by analyzing the context surrounding the Named Entity.
- In the following two sentences, the word “Amal” means two different senses:
 1. “الشباب هم أمل البلد” which means: the youth is the hope of the country.
 2. “أمل بنت جميلة” which means: Amal is a beautiful girl.



Vowels

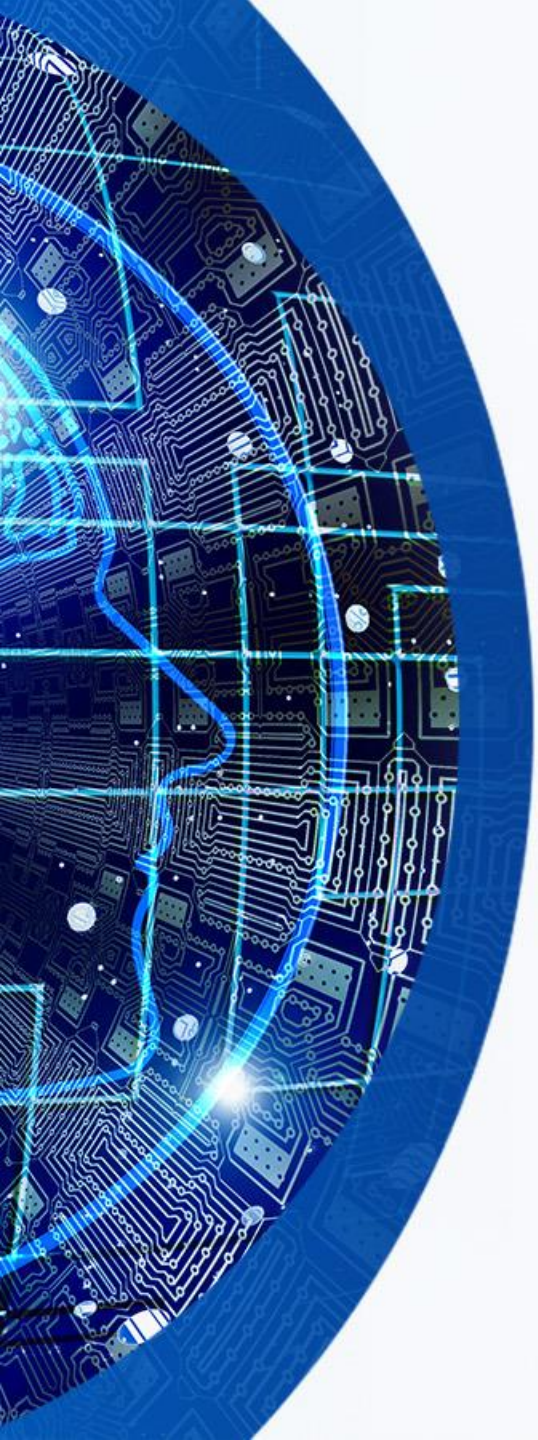
- In written Arabic, there are two types of vowels: **diacritical symbols and long vowels.**
- Arabic text is dominantly written without diacritics which leads to major linguistic ambiguities in most cases as an Arabic word has different meaning depending on how it is diacritized.

Table 3. Homographic issue for long vowels.

Word	Transliteration	Meaning	Marks
قالوا	Qalo	Said-they	No pronunciation for the letter Alef at the end
لكن	Lakin	But	No appearance of the letter Alef in the middle but pronounced

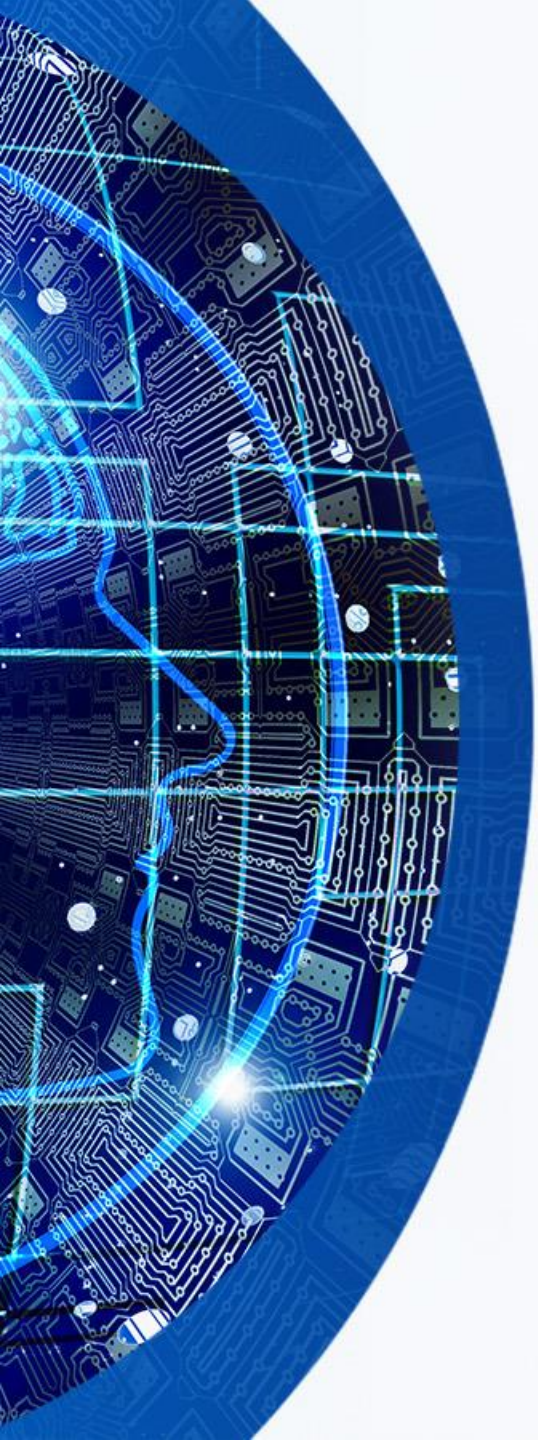
Vowels

Short Vowels and Long Vowels الحَرَكَاتُ الْقَصِيرَةُ وَالطَّوِيلَةُ					
Long /uu/	Short /u/	Long /ii/	Short /i/	Long /aa/	Short /a/
الْوَاو (و)	الضَّمَّة (ُ)	الْيَاء (ي)	الْكَسْرَةُ (ِ)	الْأَلِف (ا)	الْفَتْحَةُ (َ)
فُول	فُل	سَمِيع	سَمِعَ	جَالَسَ	جَلَسَ
Broad beans	Jasmine	All-Hearing (God)	He heard	He sat with	He sat
ثُوم	ثُمَّ	حَزِين	حَزِنَ	قَاتَلَ	قَتَلَ
Garlic	Then, after that	Sad	He was sad	He fought	He killed
لَنْ يَقُومَ	لَمْ يَقُمْ	كَفِيل	كَفَلَ	سَامَحَ	سَمَحَ
He will not stand	He did not stand	A guarantor	He guaranteed	He forgave	He permitted
عُود	عُدْ	سَلِيم	سَلِمَ	رَافَعَ	رَفَعَ
A lute, wood	Come back!	Sound, unhurt	He was safe	He pleaded	He raised
دُور	دُرْ	عَلِيم	عَلِمَ	دَارَسَ	دَرَسَ
Houses	Turn!	Knowledgeable	He knew	He studied with	He studied



Vowels

- A diacritic sign (Tashkeel Or Harakat) is not an orthographic letter.
- It is formed as diacritical marks above or below a consonant to give it a sound.
- Diacritics could also be classified into two main groups based on their functions.
- The first group includes the lexeme diacritics that determine the Part of Speech (POS) of a word as in (كتب wrote, kataba) and ,books (كُتُب kutub),

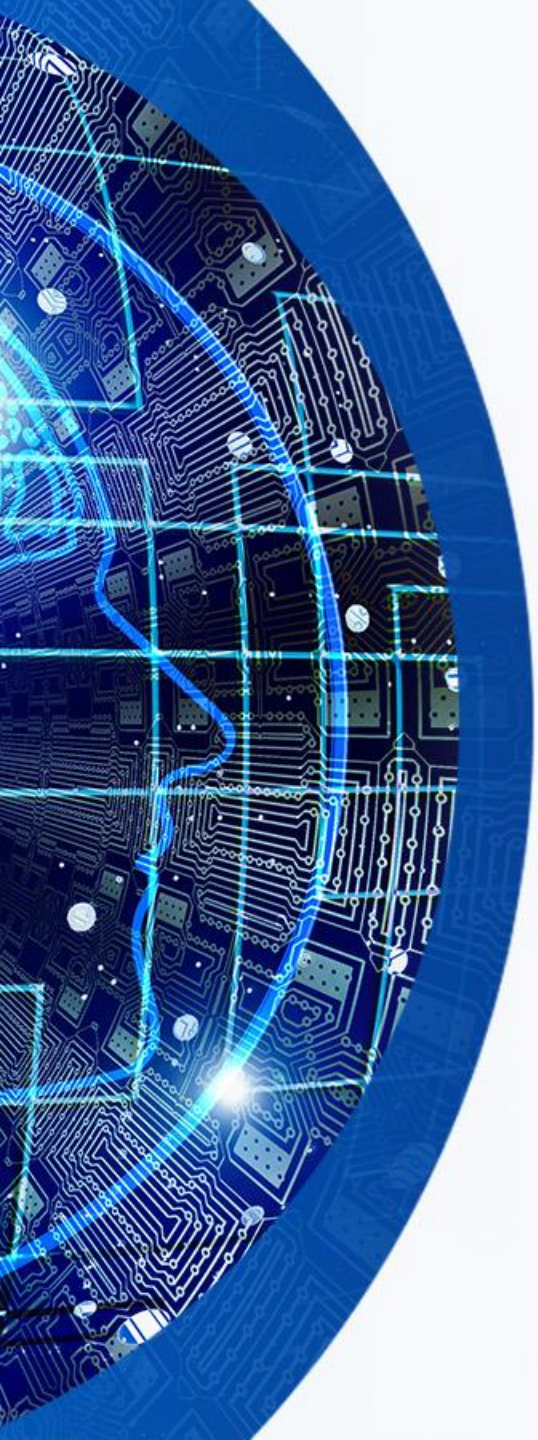


Vowels

- The second category represents the syntactic diacritics that reflect the syntactic function of the word in the sentence.
- For example, in these sentences the POS will be different:

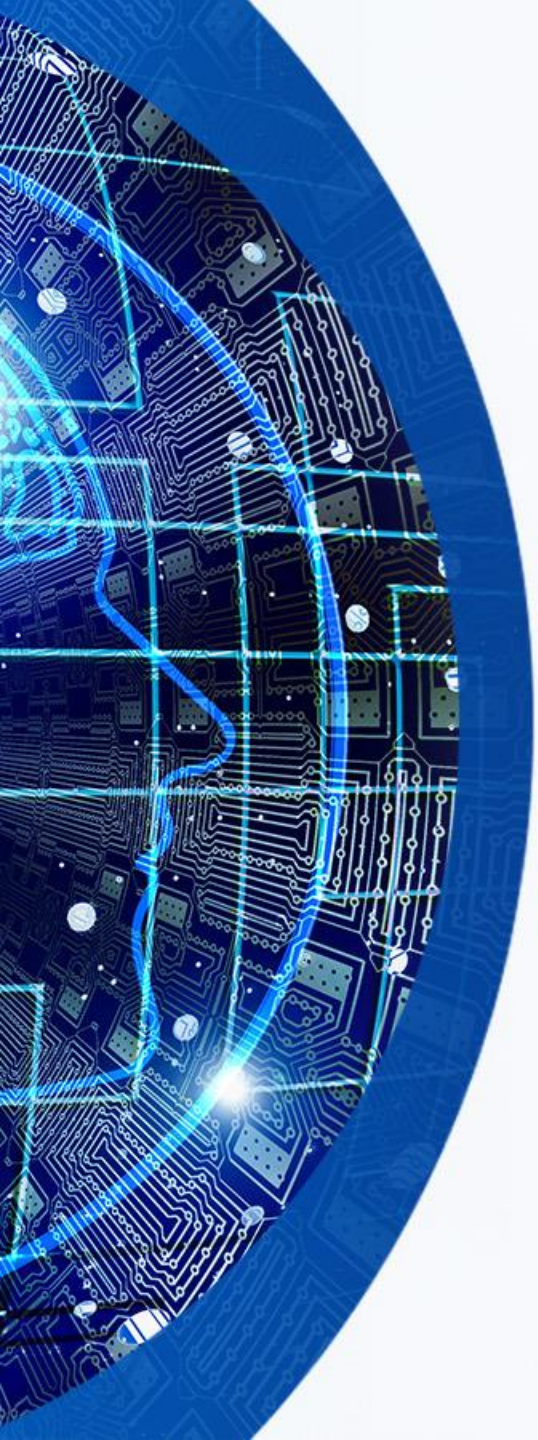
زَارَ الْوَلَدُ الْحَدِيقَةَ

تَزَيَّنَّتْ الْحَدِيقَةُ



Lack of uniformity in writing styles

- The high level of ambiguity of the Arabic script poses special challenges to developers of NLP areas such as Morphological Analysis, Named Entity Extraction and Machine Translation.
- These difficulties are exacerbated by the lack of comprehensive lexical resources, such as proper noun databases, and the multiplicity of ambiguous transcription schemes.
- The process of automatically transcribing a non-Arabic script into Arabic, is called Arabization.
- For example, transcribing an NE such as the city of Mariam Fakh ²⁹



Arabic morphology

- An additional property of Arabic that should be noted is that Arabic is an exceptionally **morphological rich lingo**.
- Its vocabulary can be easily amplified using a framework that allows for a creative use of roots and morphological samples.
- There are 85% of words from tri-demanding roots and there are around 10,000 free roots.
- Hence, Arabic is highly derivational and inflection results in high inflections in morphology
- Arabic is known for its templatic morphology where words involve roots and illustrations in the form of patterns, and fastened with affixes.

Morphology is intricate

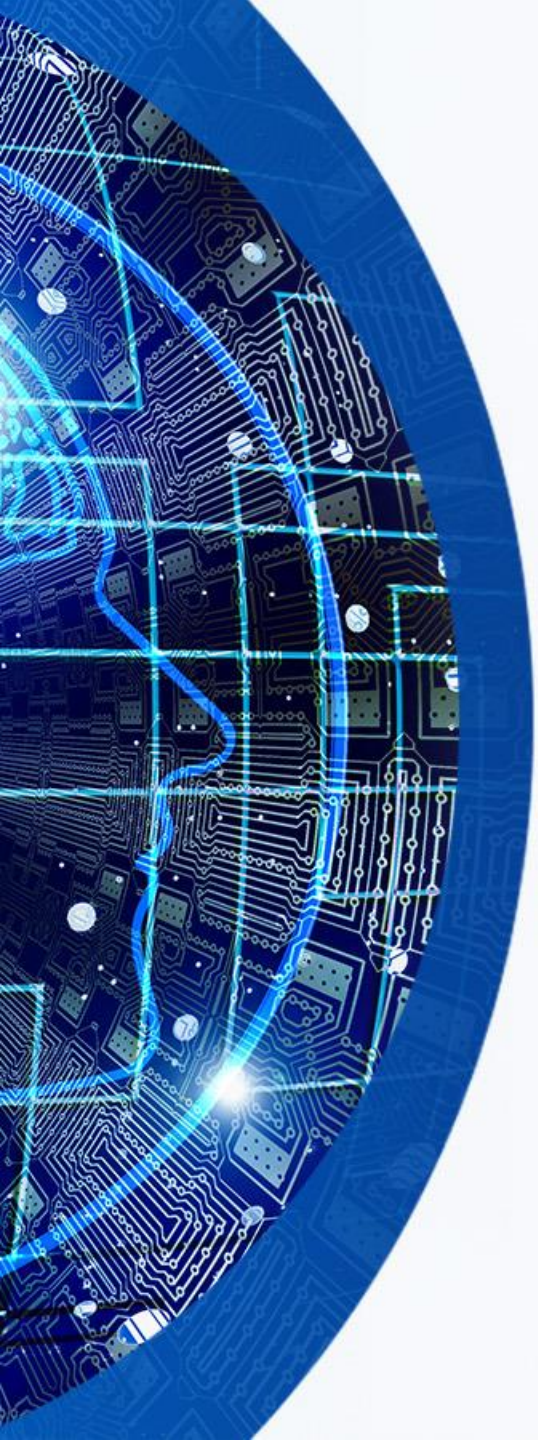
- Arabic is highly **derivational**.
- All the Arabic verbs are derived from a base of **three or four characters' root verb**.
- Essentially, every one of the descriptors gets from a verb and every one of them are inferences too.
- Deductions in Arabic are quite often templatic; hence, we can say simply that: Lemma = Root + Pattern.

Table 4. Illustration of Arabic Language in the derivational stage.

Lemma	Transliteration	= Root	Transliteration	+ Pattern
مفتوح	Maftooh	فتح	Fath	م _ ؟ و _ ؟
مدرس	Madroos	درس	Daras	

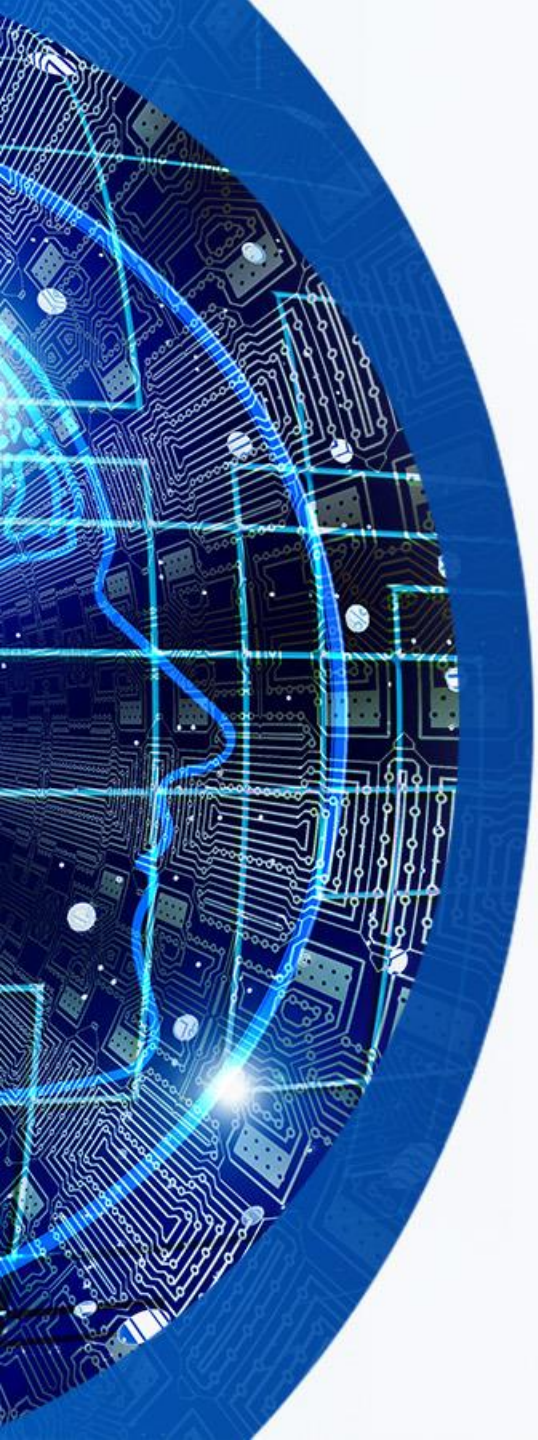
Morphology declension

- Arabic is highly inflectional.
- The prefixes can be articles, relational words or conjunctions, though the suffixes are by and large protests or individual/possessive anaphora.
- both prefixes and suffixes are permitted to be mixes, and along these lines a word can have zero or more affixes, i.e.
- Word = Prefix(es) + Lemma + Suffix(es).
- A more complicated example would be words that could represent an entire sentence in English such as “وسيحضرونها” and they will bring it, wasayahdurunaha. This word can be written in this form: وسيحضرونها = و + س + ي + حضر + ون + ها
(wa+sa+ya+hdr+runa+ha, and+will+bring+they +it)



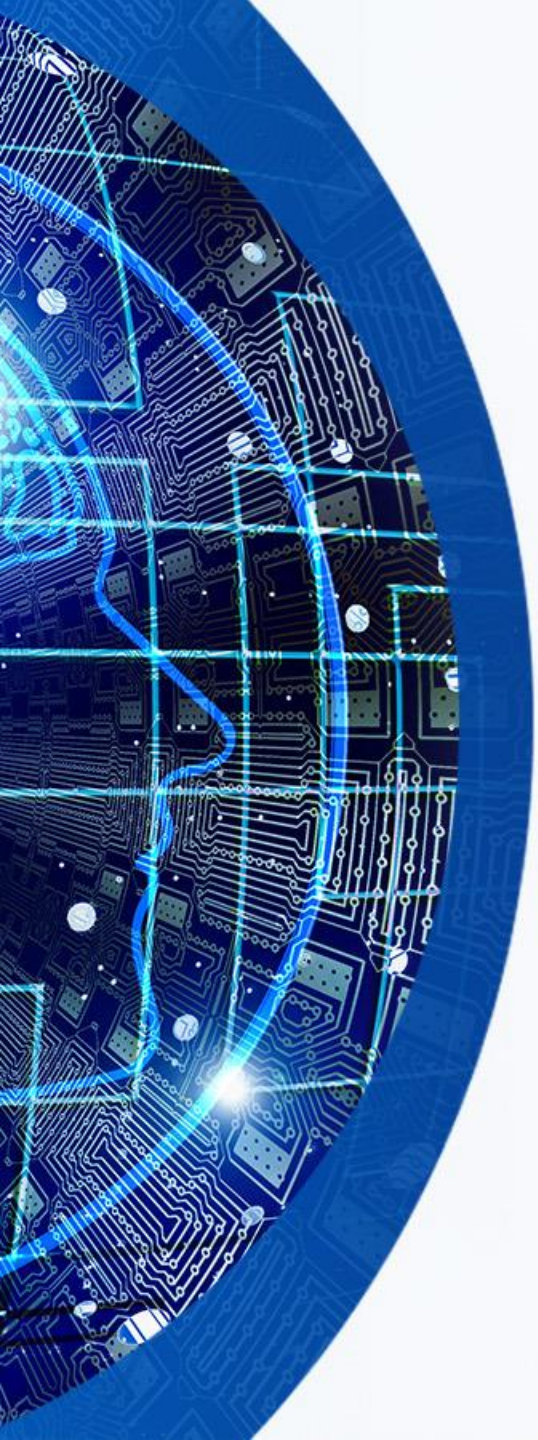
Annexation

- Another morphologic challenge in Arabic language is that we can **compose** a word to another by a **conjunction of two words**.
- This conjunction can be with nouns, verbs, or particles. Although it is **not common in traditional Arabic language**, it is used in **Modern Standard Arabic**.
- Usually, the compound word is **semantically transparent** such that the meaning of the compound word is compositional in the sense that the meaning of the whole is equal to the meaning of parts put together.
- For example, the word “رأسمالية” (capitalism, rasimalia) comes from compound of two nouns “المال رأس” (capital, ras almal)



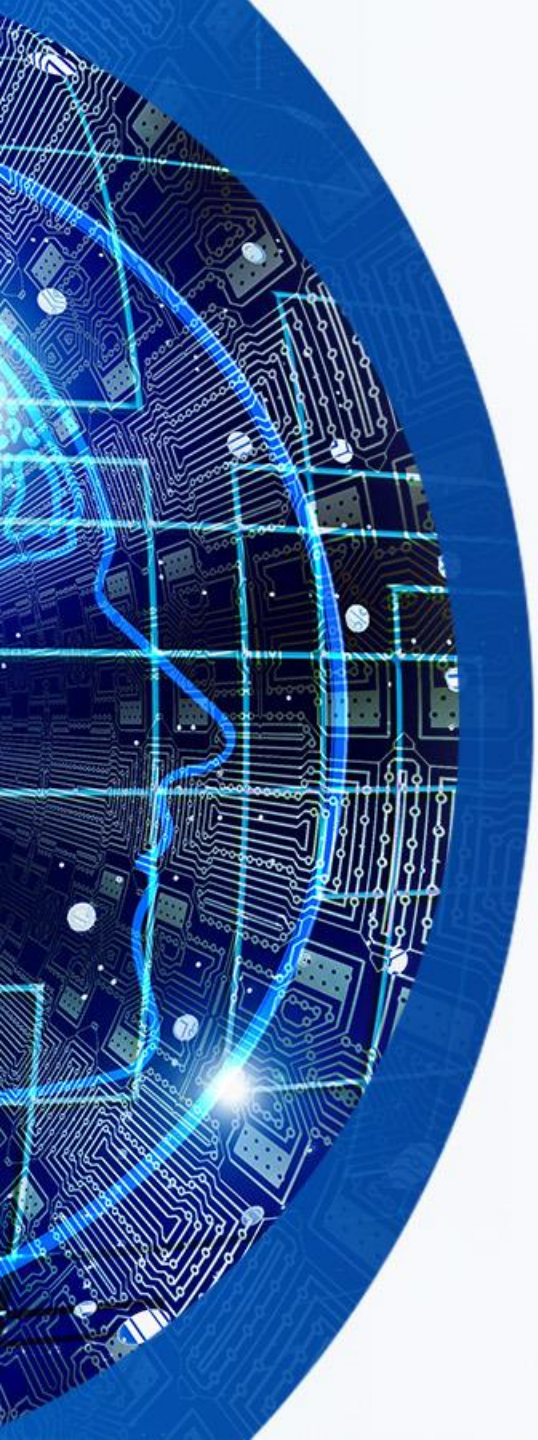
Syntax is intricate

- Arabic syntax is intricate.
- Automating the process that makes the computer analyze the Arabic sentences is truly a challenging problem from the computer perspective.
- Arabic grammar distinguishes between two types of sentences: verbal and nominal.
- Verbal sentences usually begin with a verb, and they have at least a verb (“فعل”, faeal) and a subject (“فاعل”, faeil). The subject as well as the object can be indicated by the conjugation of the verb, and not written separately.



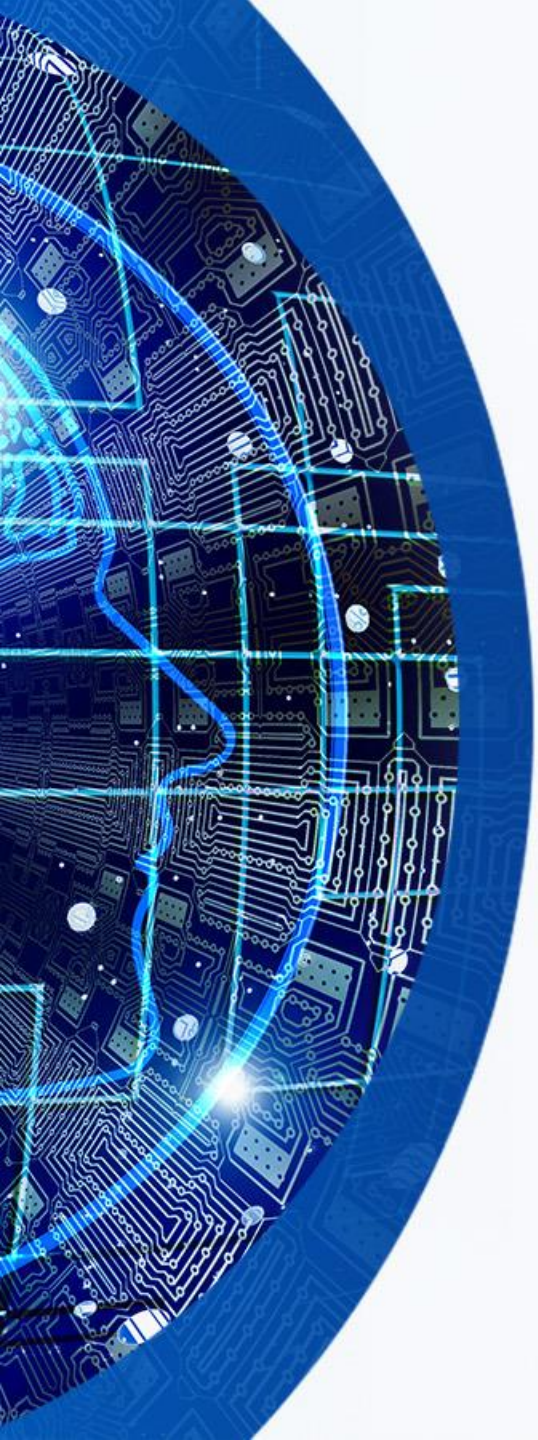
Syntax is intricate

- For example, the conjugated verb “I saw you, saw-I-you, shahidtuk) (شاهدتك “ has a subject and an object suffix pronouns attached to it.
- Another example of a verbal sentence is “الولد يدرس” “studying the boy, “yadrus alwald”.
- This type of sentence is not applied in English sentences.
- All the English sentences begin with a subject, and followed by a verb, for example, “the boy is studying”.



Syntax is intricate

- In Arabic, a nominal sentence begins with a noun or a pronoun. The nominal sentence has two parts: a subject or topic “مبتدأ”, (mubtada) an a predicate “خبر”, (khabar).
- The nominal sentences have two types: with or without a verb. The nominal verbless sentence is a typical noun phrase
- For example, “الطقس جميل” “alttaqs jamil” has two nouns without a verb; its English translation is “The Weather [is] wonderful”.
- This can be confusing to second language learners who speak European languages and are used to have a verb in each sentence



Multi word expressions

- Multi word expressions are very important constructs because their total semantics usually cannot be determined by adding up the semantics of the parts.
- For example, the multi words expression “بالحديد والنار” (by force, bialhadid walnnar) consists of two words that have the literal meaning “حديد” (iron, hadid) and “نار” (fire, nar).
- These non-decomposable lexicalized phrases are syntactically-unalterable units that are unable to capture the effects of inflectional variation.
- Thus, they can cause problems in Machine Translation, Information Retrieval, Text Summarization, among other NLP applications. Such expression is termed as idiomatic multi word expressions

Anaphora resolution

Translation: "My brother Samir has original and fake clothes."

Referent

أخي سمير عنده ملابس أصلية وأخرى مقلدة.

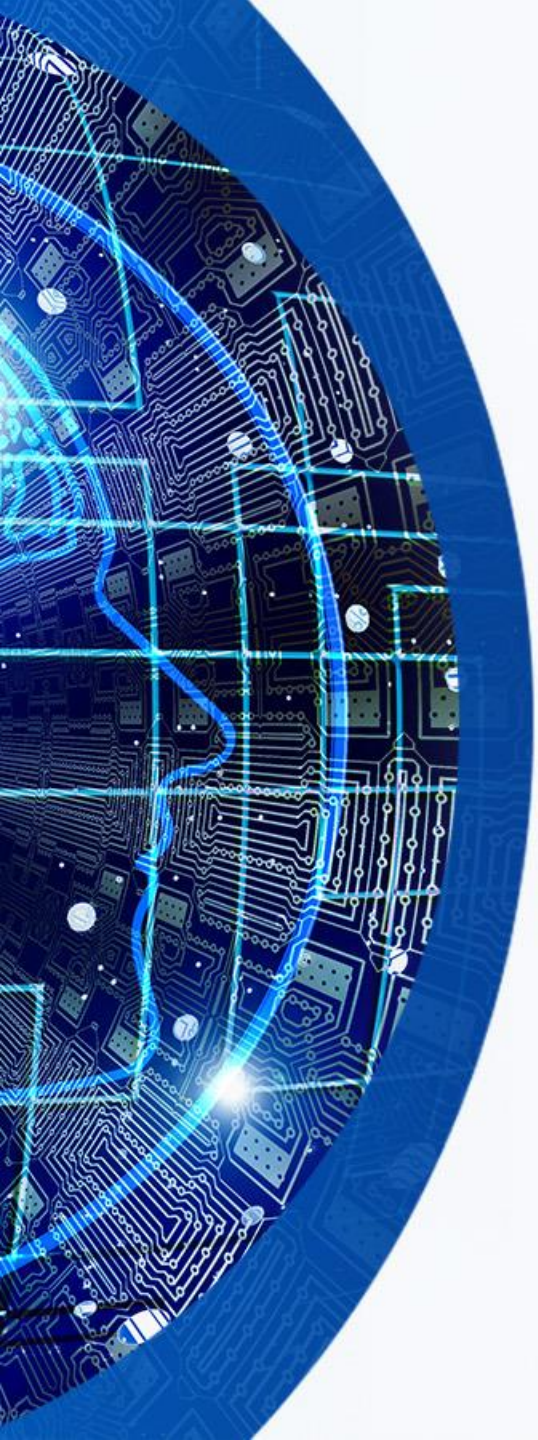
Comparative Anaphora

Translation: "the passengers have to do all the tests, otherwise they don't get on the plane."

Referent

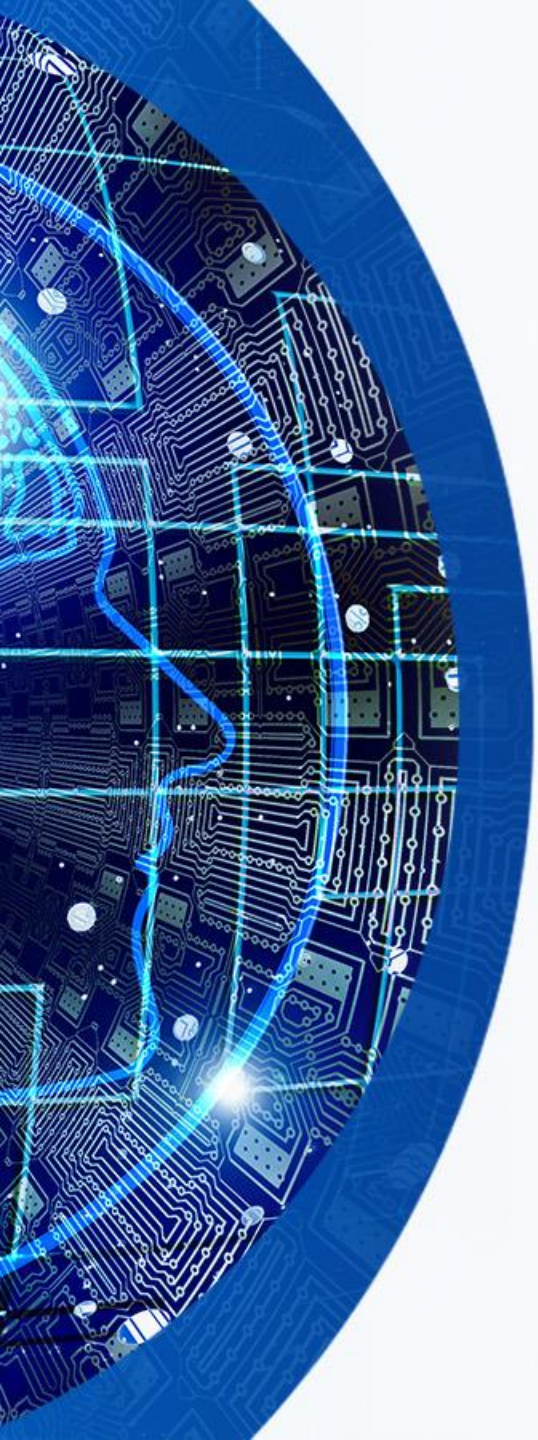
على المسافرين القيام بجميع التحاليل،
فإن لم يفعلوا فلن يركبوا الطائرة.

Verbal Anaphora



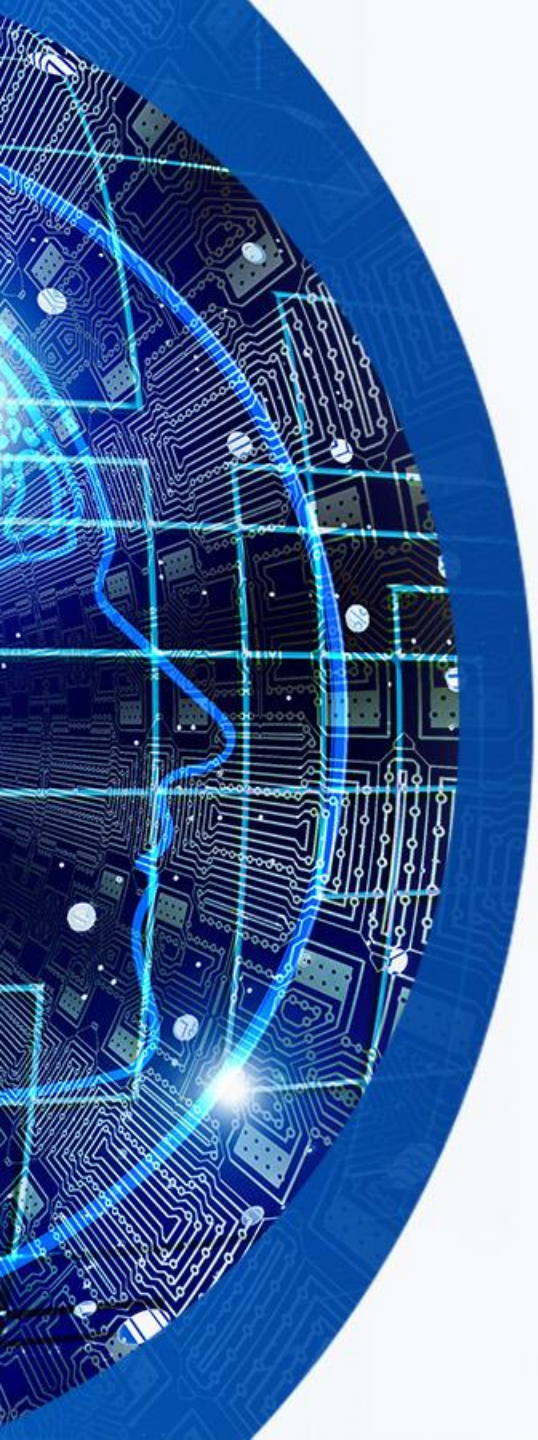
Anaphora resolution

- Anaphora Resolution is classically recognized as a very difficult problem in NLP.
- It is one of the challenging tasks that is very time consuming and requires a significant effort from the human annotator and the NLP system in order to understand and resolve references to earlier or later items in the discourse.



Ambiguous Anaphora

- The pronominal anaphora is a very widely used type in Arabic language as it has empty semantic structure and does not have an independent meaning from their antecedent; the main subject.
- This pronoun could be a third personal pronoun, called “الغائب ضمير” (damir alghayib) in Arabic,
- such as “ها” /hA/ (her/hers/it/its),
- “ه” /h/ (him/his/it/its),
- “هم” / hm/ (masculine: them/their),
- and “هن” / hn/ (feminine: them/their)



Ambiguous Anaphora

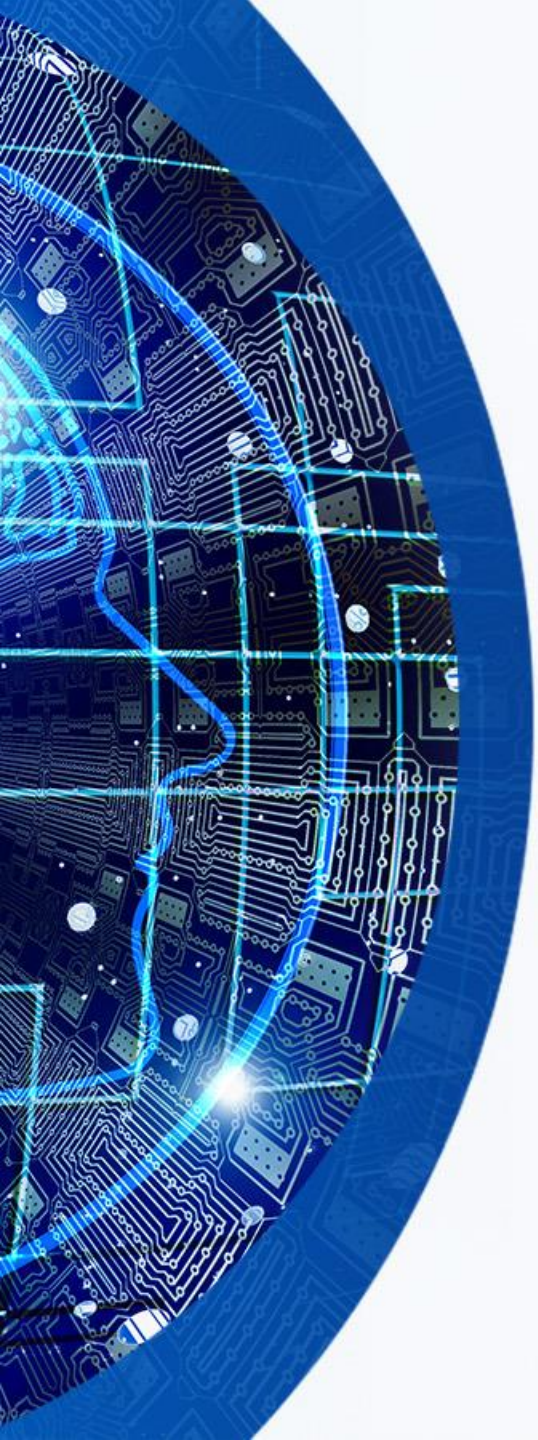
- As an example that shows the challenges of pronominal anaphora to NLP tasks, consider the result of using Google Translate© to translate two Arabic sentences into English|:

رأيت القطة ، فأعطيها الطعام

- Transliteration: ra'ayt alquttah, fa'aetiatuha alttaeam
- Google translation: I saw the cat, so I gave **her** food
- Correct translation: I saw the cat, so I gave **it** food

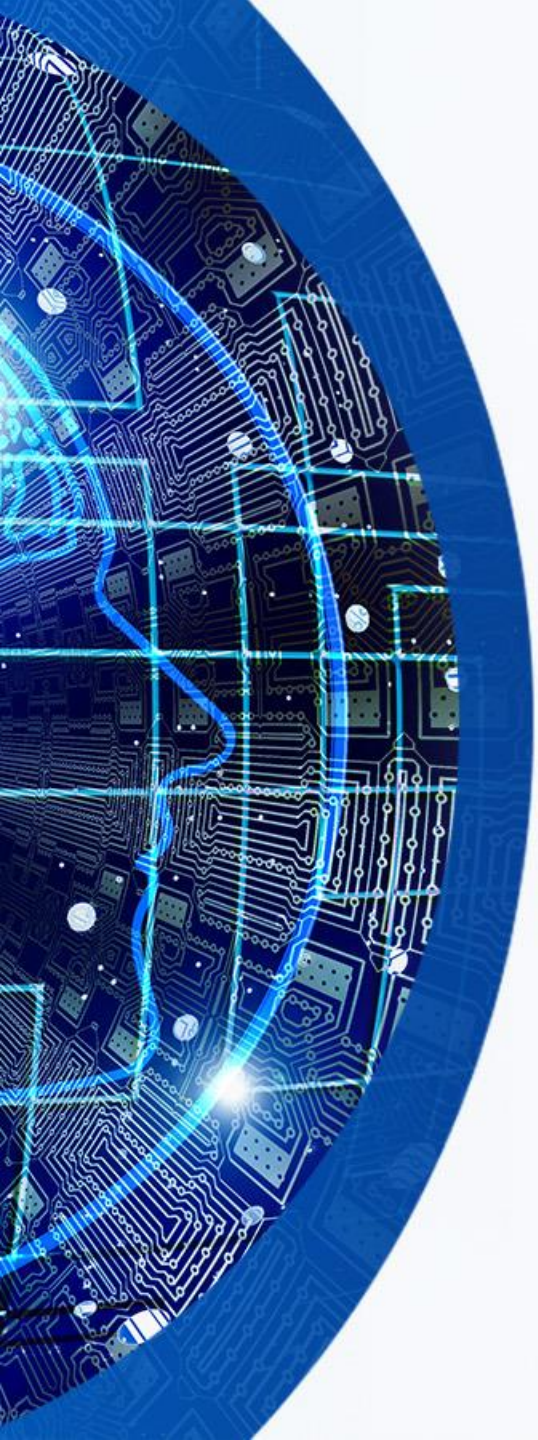
رأيت الطفلة ، فأعطيها الطعام

- Transliteration: ra'ayt alttaflah, fa'aetiatha alttaeam
- Google translation: I saw the little girl, so I gave **her** food
- correct translation: I saw the little girl, so I gave **her** food



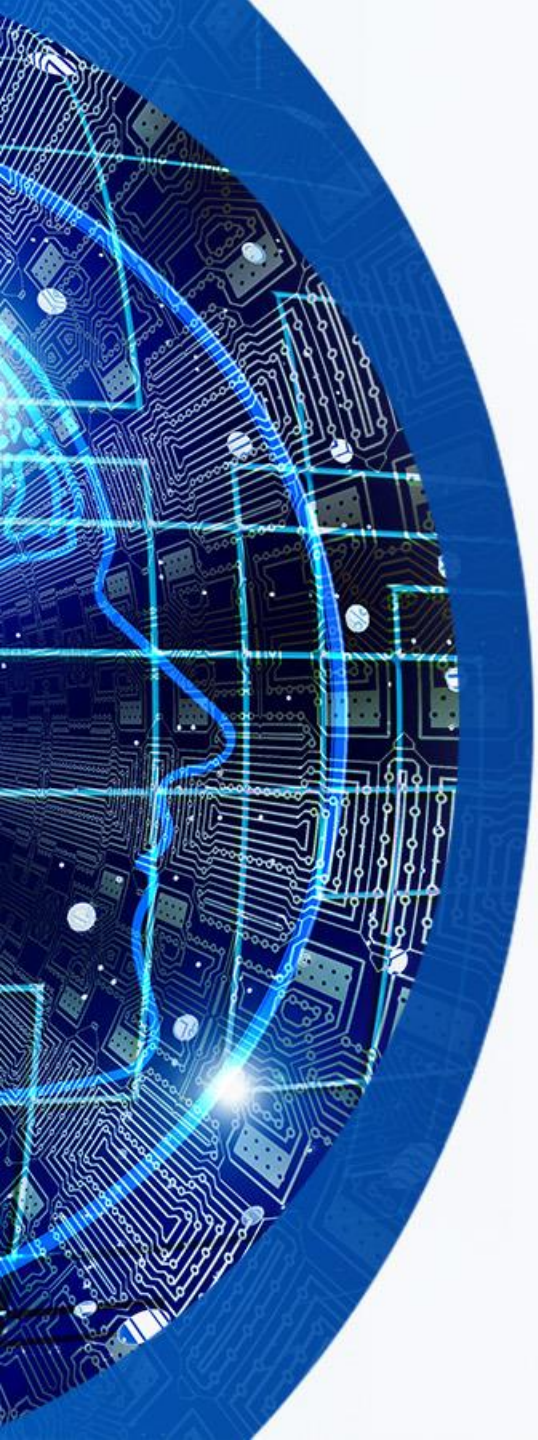
Ambiguous Anaphora

- The machine translation system fails to identify the correct antecedent indicated by the third personal pronoun “ها” /hA/ (her/hers/it/its) and thus external knowledge is needed in order to correctly identify this antecedent.
- Arabic pronominal does not differentiate linguistically between the value of the humanity feature, i.e. $\pm$ human. As a result, both the -HUMAN FEMININE noun “القطّة” (the cat) and the +HUMAN FEMININE noun “الطفلة” (the little girl), causes ambiguity in the translated English sentence.



Hidden Anaphora

- Another major kind of anaphora is hidden anaphora. It is restricted to the subject position when there is no present noun or pronoun acting as the subject.
- This is evident in the following sentence: “مَعْقِدَةٌ ، الملاحظة على اللوح” (The note on the board, complex) where the pronoun “هي” is not presented in the sentence, i.e. “الملاحظة على اللوح ، هي معقدة” , which is called “zero anaphora”.
- The human mind can determine the hidden Anaphora (antecedent) but it causes grammatical mistakes in automated NLP systems.



Syntactically flexible text sequence

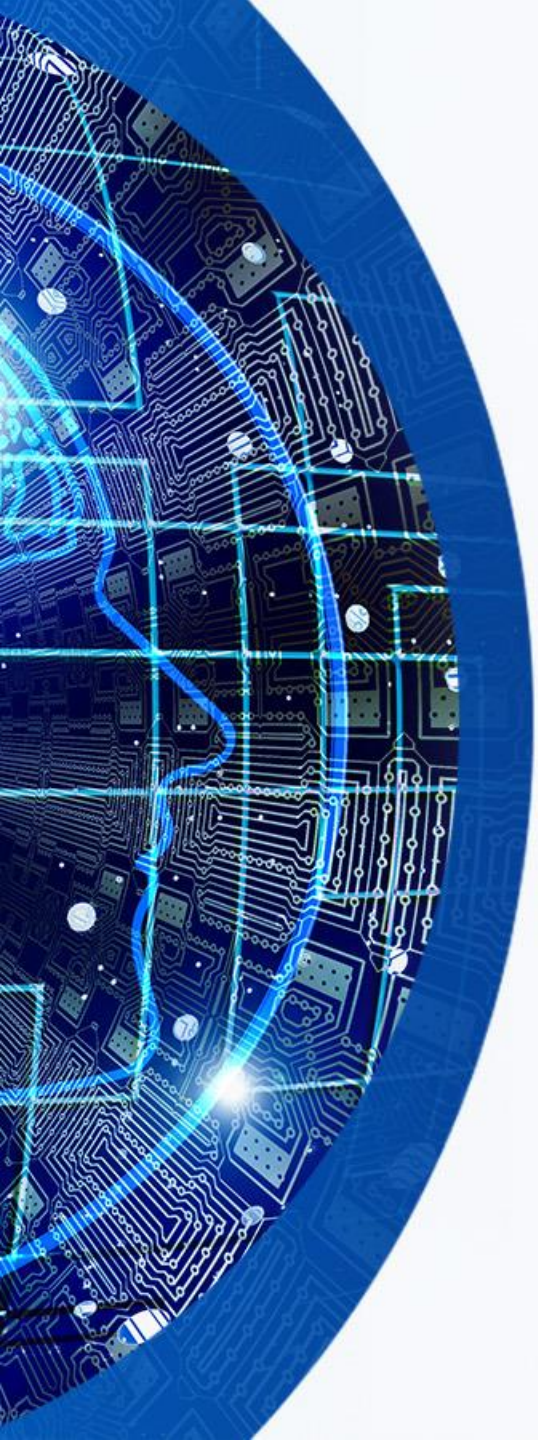
- Syntactically-flexible expressions exhibit a much wider range of syntactic variability and types of variations possible are in the form of **Verb-Subject Object** constructions.
- Arabic is generally a **free word** request language.
- While the essential word order in Classical Arabic and Modern Standard Arabic is verb subject-object (VSO), they likewise permit subject-verb-object (SVO), object-subject-verb(OSV) and object-verb-subject (OVS).
- It is basic to utilize the SVO in daily papers features.
- Arabic vernaculars display the SVO request.

Syntactically flexible text sequence

- Word order disparity is depicted in Table 5. This makes the sentence generation of Arabic NLP applications a challenge. For example, in a question-answering system, the answer to the question “أين كتاب هدى؟” (where is Hoda's book? 'ayn kitab huda?) could be any sentence that is shown in Table 5 which indicates that Huda sold the book

Table 5. Word order disparity.

Examples in Arabic	Transliteration English	Translation in English	Order
باعت هدى الكتاب	sold Huda-NOM book-ACC	Huda sold the book	VSO
هدى باعت الكتاب	Huda-NOM sold book-ACC	Huda, she sold the book.	SVO
الكتاب هدى باعته	DEF-book-NOM Huda-NOM sold-it	The book, Huda sold it.	OSV
الكتاب باعته هدى	DEF-book-NOM sold-it Huda-NOM	The book, Huda sold it.	OVS



Agreement

- Agreement is a major syntactic principle that affects the analysis and generation of an Arabic sentence which is very significant to difficult NLP applications such as Machine Translation and Question Answering.
- Agreement in Arabic is full or partial and is sensitive to word order effects.
- An adjective in Arabic usually follows the noun it modifies “الموصوف” (almawsuf) and fully agrees with respect to number, gender, case, and definiteness, e.g. “الولد المجتهد” (The diligent boy, alwald almujtahad) and “الأولاد المجتهدون” (The diligent boys, al'awlad almujtahidin).

Agreement

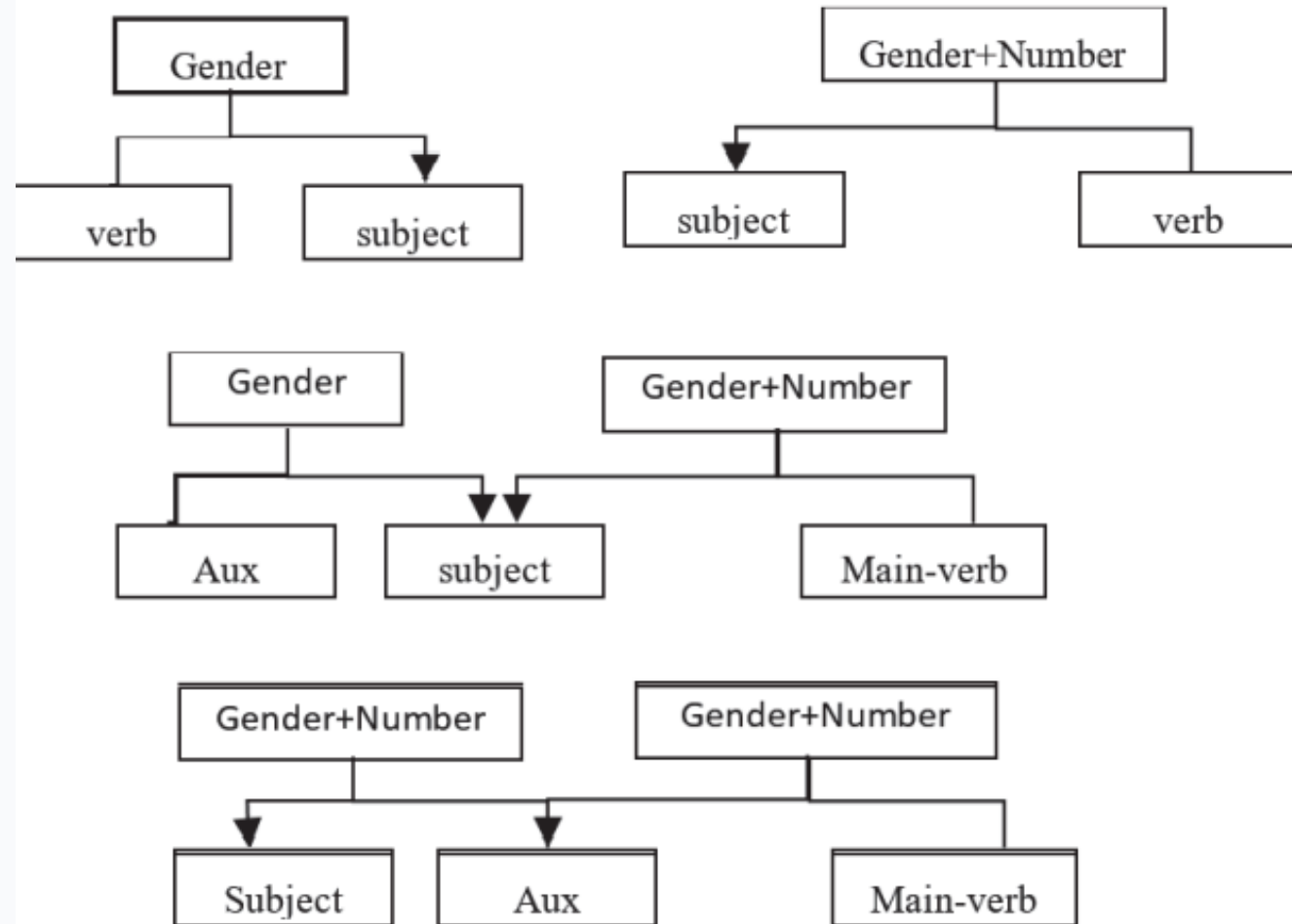
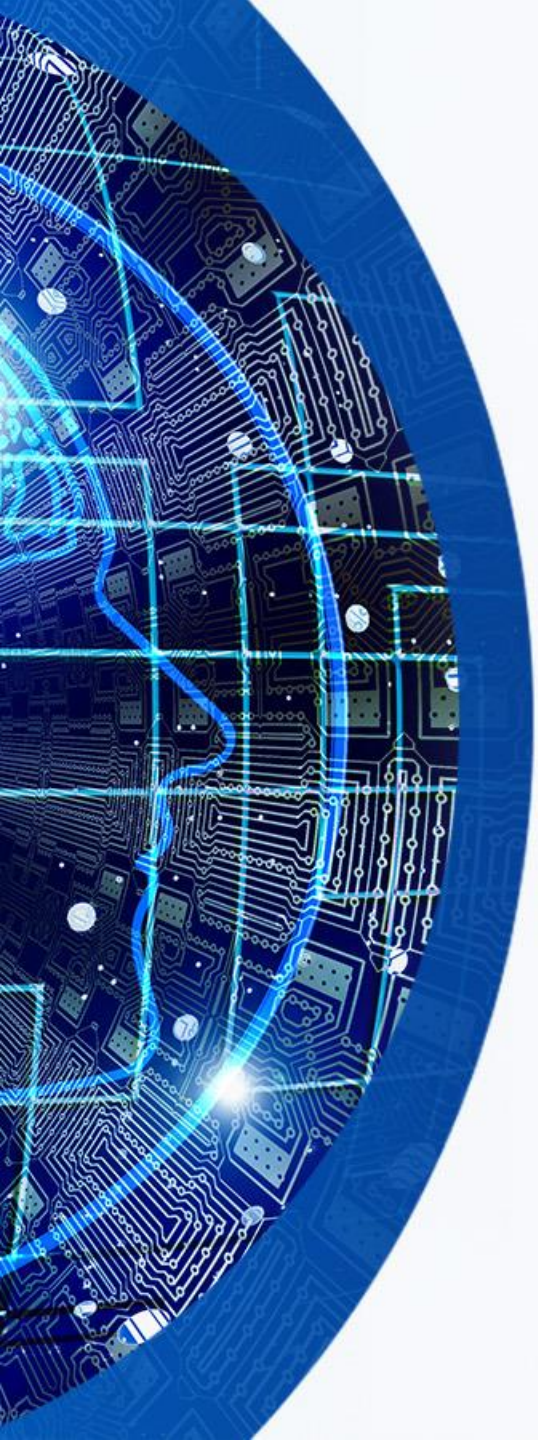


Fig. 1. Agreement patterns in verb-subject vs. subject-verb word order.



Agreement

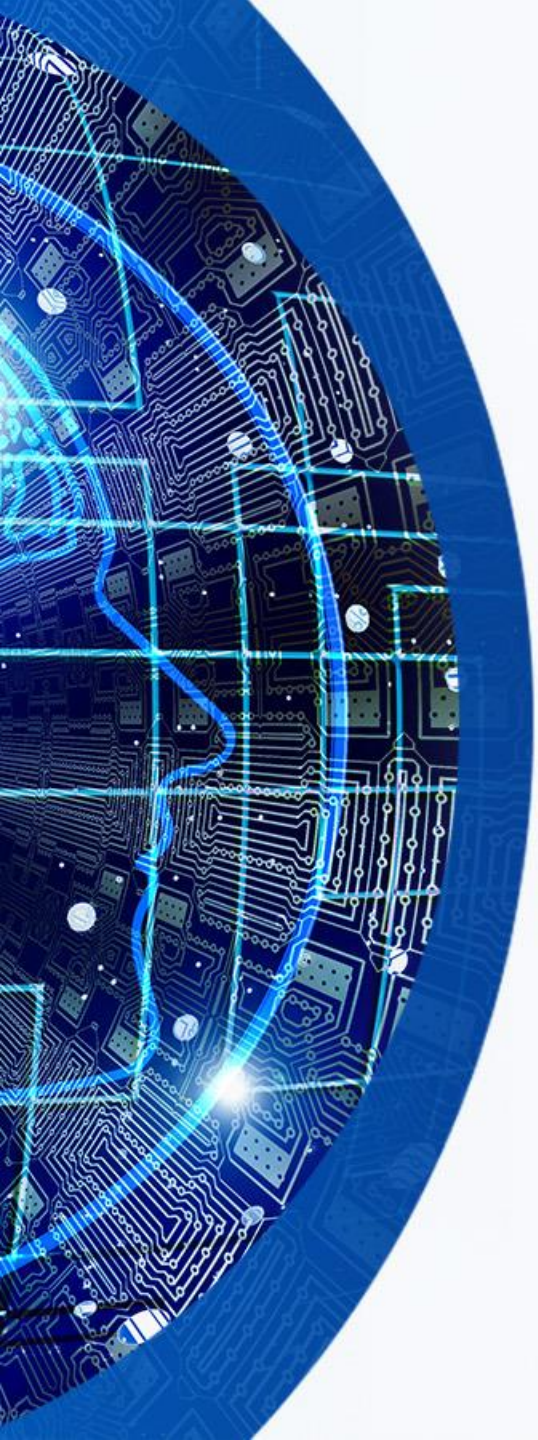
- The verb is marked for agreement depending on the word order of the subject relative to the verb
- The verb in Verb-Subject-Object order agrees with the subject in gender, e.g. “الأولاد / الولد جاء” (came the-boy/the-boys, ja' alwalad/ ja' al'awlad) versus “البنات / البنت جاءت” (came the-girl/the-girls, ja'at albint/ ja'at albanat).
- In Subject-Verb-Object (SVO) order, the verb agrees with the subject with respect to number and gender, e.g. “جاءوا الأولاد” (came the-boy/the-boys) versus “جئن البنات / جاءت ” (came the-girl/the girls)

Agreement

Number Agreement

Identify the جذر of each adjective and make it agree with its noun.

باب <u>كبير</u>	بابان <u>صغير</u>	غرفتان <u>كبير</u>	غُرف <u>صغير</u>	بنتين <u>ذكي</u>
ك - ب - ر	ص - غ - ر	ك - ب - ر	ص - غ - ر	- - - -
كبير	صغيران	كبيرتان	صغيرة	_____



Conclusion

- Arabic as a language is both challenging and interesting.
- An Arabic word does not dedicate letters to represent short vowels. It requires changes in the letter form depending on its place in the word, and there is no notion of capitalization
- Arabic Anaphora has increased the ambiguity of the language, as in some cases the Machine Translation system fails to identify the correct antecedent because of the ambiguity of the antecedent
- Moreover, Arabic sentence constituents (free word order) can be swapped without affecting structure or meaning, which adds more syntactic and semantic ambiguity, and requires analysis that is more complex.
- Nevertheless, agreement in Arabic is full or partial and is sensitive to word order effects.
- Arabic language differs from other languages because of its complex and ambiguous structure that the computational system has to deal with at each linguistic level.